MSc. Data Science Thesis

Conversational Explanations using Large Language Models on top of the eXplainable AI Question Bank

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Abstract

The proliferation and widespread adoption of AI systems into our everyday lives poses a significant risk when these systems behave unexpectedly, or make decisions which unfairly target people/groups of people. AI systems need to be transparent, interpretable, and understandable, in order for them to be trustworthy. Central to achieving this is the field of explainable AI (XAI), where Large Language Models (LLMs) present a valuable opportunity as conversational explanatory agents, transforming complex topics into accessible narratives. These LLMs offer personalised learning, refining the user's mental models, and filling in knowledge gaps.

In this work, an AI system was created as a prediction tool and explanation interface for vaccination uptake within the healthcare domain. A selection of LLMs (ChatGPT$_{5.2}$, Claude$_{\text{Opus 4.6}}$, Gemini$_{3\text{ Pro}}$) were evaluated on their ability to answer questions from the extended explainable AI question bank (XAI-QB) when provided with this AI system's source code. Using reference human answers as ground-truth labels, the LLMs were evaluated across the dimensions of *accuracy*, *completeness*, and *faithfulness*, which were used to formulate an overall ACF score. Furthermore, these reference human answers served as a baseline for *readability*, benchmarking the required level at which technical details could be presented in a comprehensible way.

The result of these evaluations suggests that LLMs can indeed be used as conversational explanatory agents. All LLMs performed comparably well, showcasing their ability to answer questions correctly, covering all of the required topics, in a semantically similar way to reference human answers. ChatGPT$_{5.2}$ performed the most consistently, ranking first in *accuracy* and *faithfulness*. Claude$_{\text{Opus 4.6}}$ provided the most *complete* responses, whilst Gemini$_{3\text{ Pro}}$ scored the best for *readability*. This showcases the relative strengths of different LLMs, suggesting that model selection could be tailored to the use-case and communication needs of different audiences.

Finally, the ACF score provides insights into where the XAI-QB and AI system can be improved —highlighting questions which need further clarification, and AI system design that lacks transparency.

Contents

1 Introduction

1.1 Background and Motivation

1.1.1 Artificial Intelligence (AI)

The idealisation of artificial intelligence traces back throughout the history of humankind. Through folklore and mythology, the anthropomorphisation of inanimate objects imbued with that of human intelligence and autonomy shows the continued pursuit to replicate what makes us distinctly "human". [66] The ability to learn and reason, think critically and abstractly, solve problems and make decisions. This collection of abilities distinguishes us from other (non)living things. [72] This pursued ideal has become much more of a reality with the advent of computers and the unabated improvements in their computational power. In modern day terms, AI is the ability of computational systems to perform tasks autonomously and emulate this human intelligence.

1.1.2 Large Language Models (LLMs)

A subdiscipline of AI, known as deep learning (DL), takes inspiration from the architecture of the human brain and its biological neural networks. Learning from huge amounts of data, it is able to make decisions and find patterns without being explicitly programmed to do so. This has led to huge advancements in areas such as computer vision (e.g. image classification, object detection), speech and audio (e.g. speech recognition, text-to-speech), and natural language processing (NLP). The transformer architecture proposed by Vaswani et al. in 2017 [84] catalysed AI research, particularly in the field of NLP. From this, emerged generative pre-trained transformer (GPT) models [68], accelerating the development of large language models (LLMs). In late 2022, OpenAI released its consumer-facing generative AI chatbot, ChatGPT, triggering an AI boom and the widespread adoption of LLMs into people's everyday lives. Since its initial release, the capabilities of LLMs have expanded greatly, with these models being trained on increasingly vast amounts of data, with a huge number of parameters. LLMs can hold long conversations, interpret and generate code, perform complex reasoning tasks, and handle multimodal inputs and outputs. LLMs have been applied to the disciplines such as medicine, scientific research, education, mathematics, finance, and law [65].

1.1.3 AI Systems

AI systems refer to the application of AI in all of its diverse forms. These systems are pervasive in our everyday lives, in varying degrees of complexity and visibility. From search engines (e.g. Google, Bing), recommendation systems (e.g. YouTube, Netflix), virtual assistants (e.g. Alexa), and autonomous vehicles (e.g. Tesla, Waymo), AI is embedded into an increasingly wide array of tools and technologies we take advantage of, and are impacted by, daily.

The European Union's (EU) AI Act [26] is the leading regulatory framework for governing how AI systems should be appropriately utilised. This is largely necessitated by the proliferation of AI tools being used, particularly in high-risk use-cases (e.g. healthcare, justice system). They define an AI system as:

> "A machine-based system that is designed to operate with varying levels of autonomy and that may exhibit adaptiveness after deployment, and that, for explicit or implicit objectives, infers, from the input it receives, how to generate outputs such as predictions, content, recommendations, or decisions that can influence physical or virtual environments." [83]

1.1.4 Human-centred AI (HC-AI)

The justification for decisions made by these AI systems are often non-existent, or obfuscated behind proprietary algorithms. As these decisions become more frequent and consequential, this raises huge concerns over the accountability and responsibility of an AI systems' judgements. Human-centred AI (HC-AI) is a research philosophy and design approach, prioritising human needs and values in AI development. HC-AI advocates to keep the control firmly with humans, where AI serves to augment and empower human capabilities, rather than to amputate and replace [78]. HC-AI sits at the intersection of many different fields, combining computer science and human-computer interaction (HCI) with human focused disciplines such as cognitive science, psychology, ethics, sociology, law, and design. Central to HC-AI are design principles focused on user agency, where AI decisions remain under human oversight and supervision (i.e. human-in-the-loop). This ensures accountability and supports the development of fairer, more transparent, and trustworthy systems. This need for transparency is increasingly reflected in regulation, from the GDPR's "right to explanation" [34] to the EU AI Act's explainability requirements for high-risk systems [27].

1.1.5 Explainable AI (XAI)

Achieving this explainability in practice requires methods that can articulate the reasoning behind an AI system's outputs, which is a central goal of explainable AI (XAI). XAI aims to enhance user trust of AI systems through increased transparency, interpretability, and understandability of these systems. This is accomplished using a suite of techniques including:

- *Intrinsic interpretability* creating transparent models by design, such as linear/logistic regression models, decision trees, and rule based methods.
- *Post-hoc methods* approximating why a model has made a decision after it has been trained, such as feature importance [29], partial dependence plots [32], LIME [71], and SHAP values [52].
- *Counterfactual explanations* asking what needs to change, in order for the model to to change its output/decision, such as questions in the Explainable AI Question Bank (XAI-QB) [48].

XAI techniques seek to explain the model at different levels of granularity, from overall global model behaviour, to local individual predictions. In doing do, these techniques aim to explain predictions, justify decisions, uncover data limitations, and detect algorithmic bias. As these systems become responsible for increasingly complicated tasks —and consequently more complex —they become harder to interpret. This signifies an accuracy-explainability trade-off that XAI tries to minimise. Bridging this gap is key to designing, developing, and deploying AI systems that are trustworthy, accountable, and widely accepted, both in high-stakes domains and in everyday life.

1.2 Research Goal

Large Language models (LLMs) are being increasingly used as conversational explanatory agents to assist in XAI, transforming complex topics and ideas into personalised, accessible narratives. The ability to iteratively ask specific questions, refine their own mental models, and fill in knowledge gaps (i.e. grounding) is an individualised learning experience that LLMs have the ability of offering. However, the outputs of these LLMs have either been restricted to templates, or are largely evaluated based on user experience and acceptance [80, 3, 86, 81, 82, 56]. New techniques are being introduced to evaluate the responses of LLMs, assessing their output in a selection of qualitative and quantitative ways [37, 15, 47, 43, 18, 20]. However, there is no unified consensus on *what* needs to be measured, nor *how*. The research goal of this project aims to bridge this gap between XAI interfaces utilising LLMs and the evaluation techniques used to assess their responses. This research proposes an *Accuracy-Completeness-Faithfulness* (ACF) composite metric with readability benchmarking. This utilises the extended XAI-QB as a structured evaluation schema for LLM generated explanations when compared against reference human answers (i.e. ground-truth labels) and their "atomised" information units (i.e. essential terms and figures).

To pursue this, an AI system was developed around a healthcare specific dataset, serving as a prediction tool and explanation interface. This simulates a real world scenario where AI-assisted decision-making is applied to a high-stakes domain. A selection of LLMs (i.e. $\text{ChatGPT}_{5.2}$ [67], $\text{Claude}_{\text{Opus 4.6}}$ [4], $\text{Gemini}_{3\ \text{Pro}}$ [35]) are then tasked with explaining this AI system, to elucidate user understanding and enhance trust. This research explores how effectively LLMs can communicate the AI system's internal architecture, results, and explanations. Thus, validating the use of LLMs as an explanation tool for communicating AI artefacts to domain experts.

Specific questions were selected from the extended eXplainable AI Question Bank (XAI-QB) [48, 79]. These questions relate to the entire framework of the AI system, including the data, output, performance, global model-wide explanations, system functionality, and user-interface (UI). Spanning the entire AI system, these questions serve as schema for assessing the LLMs ability to "understand" the AI system in its entirety (acknowledging that LLMs can merely mimic understanding, not truly possess it). This "understanding" is an important prerequisite to being able to successfully explain the AI system [50].

Using a selection of quantifiable measures, LLM generated answers are evaluated against reference human answers to the same XAI-QB questions, in order to determine their levels of *accuracy* (if an answer is correct), *completeness* (if an answer covers all of the required material), *faithfulness* (if an answer conveys the right meaning and is logically consistent), and *readability* (how easy the answer is to understand). These four dimensions are all crucial to ensure that factually correct and relevant information is being presented in a clear and understandable way. These are important characteristics of an explanation [24, 54, 85, 64]. A composite metric combining accuracy, completeness, and faithfulness is formulated into an ACF score, which evaluates the overall quality of an LLM's answer. This measure can then be used as a comparison across other LLMs and applications. Furthermore, reference human answers set the benchmark for readability which LLMs are evaluated against, ensuring the information is accessible and understandable to target domain experts.

There are several additional outcomes that arrive from this evaluation, beyond validating LLMs as a conversational explanatory agent. The ACF score can be used to highlight questions in the XAI-QB which need further clarification and refinement. This may be due to a lack of specificity, or an open-endedness to the question which allows for interpretation. The quality of an answer is strongly correlated with the quality of the question [89]. An ill-defined or vague question therefore opens up unwanted possibilities of confabulation and hallucinations. This analysis can be used to further improve the XAI-QB, highlighting specific questions where detail can be included, tailored specifically to the AI system and its use-case. On the other hand, this open-endedness also allows for answers to extend beyond the human reference, and identify gaps in knowledge. This provides valuable insight into additional areas of exploration and use-cases for the AI system. Finally, this measure can be used to inform better design practices. Questions which are misinterpreted, or answers which are factually correct, but irrelevant, indicate the AI system is not designed in a clear or coherent way. This can be used to inform better practices and clearer AI system design.

In summary, this research aims to provides a novel approach to evaluating LLMs as conversational explanatory agents through:

- The ACF composite metric which unifies accuracy, completeness, and faithfulness into a single reproducible quality score.
- Readability benchmarking against reference human answers to determine the desired balance of reading ease and technical detail.
- Reference human answers "atomised" into information units to measure of completeness of LLM generated responses.
- Utilising the extended XAI-QB as an evaluation schema for assessing the LLM's ability to explain the entire scope of an AI system.
- Actionable feedback loops for further XAI-QB refinement and AI system design.
- A reusable instructional template —designed to be fine-tuned and altered —for prompting LLMs as conversational explanatory agents.

This framework for the evaluation and comparison of LLM generated responses is transferable across different LLMs, AI systems, domains, and use-cases.

2 Related work

2.1 Human-centred AI (HC-AI)

Over the past few years, there has been a continuous rise in the use of AI systems, with these systems claiming to achieve increasingly complicated tasks, in increasingly consequential domains. However, there is also a global bias in trust towards these AI systems, largely rooted in their unfamiliarity and lack of transparency. This is especially apparent when the decisions of these systems provide unexpected results, or negatively impacts an individual/group of individuals. This beckons major questions surrounding trustworthiness, ethics, accountability, and responsibility. When AI systems fail, this needs to be understood [73].

Human-centred AI (HC-AI) seeks to reframe the currently technology-centric landscape by shifting focus, and therefore ownership, to humans. By placing humans-in-the-loop (at the forefront), these critical decisions are under the responsibility of the people wielding them, keeping users in control and accountable for the AI-assisted conclusions they make [78].

Humans view the world through categories, distinguishing between the animate and the inanimate, what is human versus what is non-human. AI represents a boundary condition, able to convincingly mimic the uniquely "human" capabilities of intelligence and creativity. This has enabled the automation of progressively complex tasks. However, there is the misconception that with an increase in computer autonomy, there is also the unavoidable degradation of human control. HC-AI provides the framework for effective design supporting the human user, where AI implementation enhances people's capabilities, rather than replacing them. Contrary to the previously held notion that human control and computer autonomy are diametrically opposed [77], HC-AI fosters a symbiotic relationship between these two, arguing that there is no artificial intelligence without intelligence augmentation. The research by Shneiderman [78], proposes a governance structure to bridge the gap between ethical principles of AI, which are often vague, lofty targets, and its practical implementation:

Teams such as researchers and AI practitioners creating *reliable* systems.

- AI4People [31] is an initiative bringing together a committee of experts across different domains to develop an ethical framework for AI. This framework investigates the opportunities, risks, principles, and recommendations needed to build a "Good AI Society."
- eXplainable AI Question Bank [48, 79, 23] is a set of prototypical questions designed to help AI practitioners evaluate and improve the explainability of AI systems across different stages of design, development, and deployment.
- Microsoft: AI Fairness Checklist [53] is a practical checklist to help mitigate fairness related concerns throughout the AI development lifecycle.
- Microsoft: Responsible AI Toolbox [58] is an integrated collection of tools and functionalities that operationalises Responsible AI. This includes model debugging, fairness assessment, and interpretability techniques.

Organisations encouraging a culture of *safety*.

- Google: People + AI Guidebook [36] is a guide from Google's PAIR team, which offers research informed practices for designing human-centred AI products.
- Microsoft: Responsible AI [57] is Microsoft's overarching responsible AI programme and commitment to human-centredness. They outline their six core principles as: fairness, reliability, privacy and security, transparency, accountability, and inclusiveness.
- IBM: AI Ethics [42] is IBM's ethical AI initiative, establishing guiding principles and governance practices to ensure AI systems are trusted. This is achieved through their five pillars of trust: explainability, fairness, robustness, transparency, and privacy.

Industry through regulation and oversight ensuring AI is *trustworthy*.

- EU: AI Act [26] is leading regulatory framework, classifying AI systems by risk level and setting binding requirements for transparency, safety, and human oversight.
- EU: Ethical guidelines for trustworthy AI [39] are guidelines published by the EU's High-Level Expert Group on AI, defining seven key requirements that AI systems must meet to be considered trustworthy:
  - Human agency and oversight
  - Technical robustness and safety
  - Privacy and data governance
  - Transparency
  - Diversity, non-discrimination and fairness
  - Societal and environmental well-being
  - Accountability

Human-centredness promotes this idea of multi-disciplinary collaboration from the ground up. All stakeholders (e.g. AI practitioners, domain experts, data subjects) need to be considered in the design, development and deployment to ensure these AI systems are reliable, safe, and trustworthy.
In practice however, these goals are still challenging to operationalise [10]. Trustworthiness is arguably a human concept —the "hope" or "belief" in the surety of something, that it is reliable and acts as expected. As a result, this is hard metric to quantify. HC-AI defines trust as an AI system underpinned by its ability to be both accurate and explainable. That is to say, the AI system should be faithful to the ground-truth (i.e. what is *known* to be true), and it can provide a narrative (both globally and locally) as to how it came to these decisions. This definition can also be extended to include dimensions of transparency and interpretability —providing access to all of the information regarding an AI system and its ecosystem (e.g. model, data), with visibility over its inner mechanics (e.g. features, weights). HC-AI encompasses design principles that focuses on the human experience as a user, promoting consistent interfaces, accommodating evolving user needs, and providing informative feedback.
This research project implements HC-AI design principles through the creation of reference human answers for the XAI-QB, and the manual evaluation of LLM generated responses. These are used to evaluate

the component metrics of the ACF score, ensuring humans are kept in-the-loop and accountable. This human oversight is crucial when deploying LLMs as conversational explanatory agents into high-risk domains such as healthcare.

2.2 Explainable AI (XAI)

The EU AI act defines high-risk domains in Article 6, Annex III [25], stating that any AI system in the following fields are by default considered high-risk: biometrics, critical infrastructure, education, employment, essential services, law enforcement, migration, and justice. These AI systems are also commonly referred to as “black boxes”, in which the internal mechanics of the system are largely unknown, and only the inputs and outputs are observable. In contrast, “white box” systems offer complete transparency to their inner components and logic [6].

With the continued proliferation of AI systems being adopted into our everyday lives —increasingly in high-risk domains —many recognise that alongside this is the necessity to incorporate explanation features to understand such AI systems. Adjacent to this surge in AI is the field of XAI, which aims to elucidate user understanding and demystify these opaque, “black-box” algorithms.

The act of explaining is inherently human, taking place solely in the mind of the individual, each with their own mental models and priorities. Explanations require cognition, intention and contextual awareness, all of which are distinctly “human” traits. Effective explanations need to be continuously communicated and clarified using a combination of reasoning with language and visualisations.

XAI stands at the intersection of social sciences and human computer interaction, aiming to achieve understanding, acceptance and ultimately trust of an AI system through its explainability.

Whilst explainability —the process of an AI system to provide human-acceptable justifications for its predictions/output —is the core metric, there are many other dimensions to consider when evaluating an AI system, many of which overlap and interact with one another to provide a holistic view. Arguably the most essential of these dimensions is that of understandability. This evaluates the ease a user has in grasping the overall functioning and rationale of the AI system. Both transparency and interpretability are closely linked to this, whereby the inner workings and decision-making process are accessible to the user, and they are presented in a way that can be easily understood. Equally, comprehensibility indicates the capability of the audience to understand knowledge contained within the AI system.

As AI systems become more complex —and therefore inherently more opaque —it poses an accuracy-explainability trade-off, whereby performance comes at the cost of interpretability. Transparency has become a growing demand by stakeholders involved with these AI systems (either directly or indirectly). Particularly in critical contexts, users need explanations to justify these decisions, and the transparency to do so. Without interpretable, tractable and trustworthy techniques in place, this poses an insurmountable barrier to their overall adoption.

In an effort to overcome these challenges, XAI provides a growing collection of machine learning techniques, enabling humans to understand these AI systems, and optimise the complexity trade-off —increasing explainability whilst maintaining (or even improving) performance. This also indicates the way in which understanding can be partitioned:

- *Model* understandability takes into account the comprehensibility of the AI system.
- *Human* understandability considers the users cognitive predispositions.

Barredo Arrieta et al. [6], identify the audience as a cornerstone of XAI. The main goals required of XAI depending on the target audience, including dimensions of trustworthiness, causality, transferability, informativeness, confidence, fairness, accessibility, interactivity, and privacy-awareness. Recognising the audience as a vital role in explainability, human-centredness is also a pursued goal in in the context of XAI. Dwivedi et al. [24], divides the target audience (i.e. stakeholders) based on where they're situated in the development and deployment process of an AI system, recognising that the best suited XAI method is therefore relative to both the stakeholder and phase. Developers, theorists and data scientists are involved in the *understanding* phase, where new knowledge and scientific insight is pursued. This involves building robust models (iteratively debugging and enhancing) and detecting any bias in the underlying data. Users, consumers, businesses and regulators are engaged in the *explaining* phase, which occurs after the AI system has been deployed. Understanding the varying levels of contextual and psychological needs of these stakeholders are imperative for successful XAI. Miller [59], conducted extensive research into the current outlook on the explanation process through the lens of philosophy, psychology and cognitive science, providing insight into how people employ their own personal cognitive biases and social expectations. The research from [59] places XAI at the centre of AI, HCI and social sciences. Resolving the current blind-spot AI practitioners have when using XAI techniques.
Explanations can be categorised as:

- *Contrastive*: People tend to ask why an event happened instead of another
- *Selected*: Influenced by certain cognitive biases, people often only select a subset of causes as sufficient explanation
- *Social*: Explanations are a conversation/interaction where knowledge is transferred from explainer to explainee

Furthermore, it is also notable that causal explanations are more satisfying to users than probabilities or statistical generalities. All these points converge to the conclusion that explanations are contextual, whereby the explainee considers only a selection of context-relevant causes, and the explainer selects a further subset of these causes based on different criteria.
Through a literature review by Adadi and Berrada [1], it was concluded that the need for XAI can be categorised into four main motivations. XAI is primarily used to:

- *Justify* decisions, ensuring auditable, fair and ethical outcomes
- *Control* against vulnerabilities and flaws (identify and correct errors)
- *Improve* the AI system (to improve one must first understand)
- *Discover* new insights (knowledge around the system and wider ecosystem)

Furthermore, Mohseni et al.[62] classify target users of XAI into *AI novices*, *data experts*, and *AI experts*. The authors recognise these classes of users will have largely have distinct design goals for the XAI system, relating to their own use-case and motivations. AI novices desire algorithmic transparency, user reliance, trust, bias mitigation, and privacy awareness. Data experts want the ability to visualise, inspect, select, and tune the model. Lastly, AI experts are more concerned with model interpretability and debugging.
Explainability can be segmented based on several different elements. "White-box" models, which are interpretable and transparent by design, require explanations through ante-hoc methods. These methods are model specific and include *simulatability* (ability of a model to be "simulated" or thought about by a human), *decomposability* (ability to explain each and every part of a model), and *algorithmic transparency* (ability of the user to explain the process followed by the model to produce any given output from its input data). "Black-box" models on the other hand, require post-hoc explainability methods to be applied, which occurs after a model has been trained and the output has been generated. These are much more prevalent in XAI literature, due to the desirability of more performative, complex models. Post-hoc methods are generally model agnostic, and can be further grouped into *visualisation* (e.g. surrogate models, partial dependence plots (PDPs), individual conditional expectation (ICE) [33]), *knowledge extraction* (e.g. rule extraction, model distillation), *influence methods* (e.g. sensitivity analysis, layer-wise relevance propagation, feature importance), and *example-based* (e.g. prototypes and criticisms, counterfactual explanations) techniques. Post-hoc interpretability techniques are analogous to the human way of explaining decisions, with the example-based method inspired by the cognitive science of human reasoning. Across these methods, the scope of explainability can performed on either a global or local level, explaining overall model behaviour or individual predictions respectively.
For this research project, the AI system produces XAI artefacts which evaluate ("black-box") model performance, and its global explanations. These serve as the input for LLMs acting as conversational explanatory agents, which are then tasked with explaining them to non-technical healthcare practitioners. These LLMs are evaluated on their ability to communicate these artefacts in a factually complete and accessible way.

2.2.1 Evaluating Explainability

Doshi-Velez and Kim [21] outline a taxonomy of XAI evaluation techniques:

- *Application-grounded evaluations* require human experiments, typically involving domain experts within a real world application.
- *Human-grounded evaluations* employ lay users engaged in simpler human experiments that maintain the essence of the real world application.

These are both human-centred methods which can evaluate explanations both qualitatively and quantitatively. Either through subjective survey scores (e.g. usefulness, satisfaction, confidence, trust), or objective measures of behavioural, physiological, and task performance (e.g. accuracy, response time, ability to detect errors).

- *Functionality-grounded evaluations* assesses the quality of an explanation by some formal "proxy" definition, providing objective assessments without the need for time consuming and costly human studies.

Both human-centred and computational/automated methods for evaluating explanations are necessary, as there is no universal definition as to what an explanation exactly is. This will depend on the user, their use-case and the domain they are operating within. Thus, the quality of an explanation is dependent on the ability of an explanation to effectively relay a specific aspect of an AI system, the contextual relevance, and the efficacy of this on the user's own understanding.

The research by Markus et al. [54] views explainability as an overarching term, consisting of different properties and sub-properties. They determine that an explanation should be understandable to humans (i.e. interpretable through its sub-properties of clarity, broadness, and parsimony), and accurately describe the model behaviour (i.e. fidelity through its sub-properties of completeness and soundness). The authors also distinguish the different forms of explainability, which can be either *model* (i.e. using a model to explain the original. "task" model), *attribution* (i.e. rank/measure the explanatory power of a model's input features), or *example* based (i.e. selecting/creating instances from the dataset).

Furthermore, Zhou et al. [92] cross tabulate these properties and explanation types to comprehensively categorise quantitative XAI evaluation methods available in current literature. Attribution based explanations have the most evaluation techniques, with the quality of explanations largely assessed through their properties of parsimony/simplicity and soundness. The authors call for the combination of human-centred and functionally grounded of explanations to achieve a holistic and comprehensive view of explanation quality. Furthermore, they advocate for the quantitative evaluation and comparison of functionally grounded explanation types to determine the most suitable for a defined objective.

Vilone and Longo [85] lay out such comparisons between 16 quantitative explanation methods such as SHAP, LIME, integrated gradients, and Grad-CAM. Sourced from scientific articles proposing comparative approaches to evaluate methods for explainability, these comparisons can be categorised in three main ways,: *sensitivity to input perturbation* (input feature removed/altered/masked), *sensitivity to model parameter randomisation* (some/all features replaced with random values for model training), and *explanation completeness* (which method describes the inferential process to the highest extent). The authors also curated a collection of 14 qualitative and 19 quantitative methods for evaluating explainability, through the use of human studies with either open or close-ended studies respectively. This human-centred evaluation can be of several difference construction approaches (e.g. ML (model agnostic), neural networks, context aware systems), and input formats (e.g. textual, numerical, visual, mixed).

In an effort to provide researchers/practitioners with techniques to thoroughly evaluate, benchmark and compare XAI methods Nauta et al. [64], developed the Co-12 properties based on their analysis of over 300 XAI related research papers. Arguing that explainability is a multi-faceted concept, these properties identify the major conceptual aspects which assess the quality of an explanation. Similar to Zhou et al. [92], this acts as as a categorisation scheme for quantitative XAI evaluation methods, and forms the basis for optimisation criteria when developing an XAI system.

Category	Property	Description
Content	Correctness	How faithful the explanation is w.r.t. the black box.
	Completeness	How much of the black box behaviour is described in the explanation.
	Consistency	How deterministic and implementation-invariant the explanation method is.
	Continuity	How continuous and generalisable the explanation function is.
	Contrastivity	How discriminative the explanation is w.r.t. other events or targets.
	Covariate complexity	How complex the (interactions of) features in the explanations are.
Presentation	Compactness	The size of the explanation.
	Composition	The presentation format and organisation of the explanation.
	Confidence	The presence and accuracy of probability information in the explanation.
User	Context	How relevant the explanation is to the user and their needs.
	Coherence	How accordant the explanation is with prior knowledge and beliefs.
	Controllability	How interactive or controllable an explanation is for a user.

Focusing on functionally grounded evaluation techniques, the authors identified 29 automated, quantitative evaluation methods for XAI techniques, with most methods focusing on properties of correctness and completeness.

Hoffman et al. [40] advocate for a multi-method approach when evaluating XAI, identifying four major classes of measures. The "goodness" of explanations (a priori goodness checklist for researchers), user satisfaction of these explanations (posterior satisfaction scale for users), how well users understand the AI system based on their own mental models (elicitation tasks), and how well the human-AI system performs (in regards to: primary task goals, the user, and the work system). Furthermore, they construct a trust scale to determine the user's confidence in the XAI system and gauge the user's perception of the systems predictability, reliability, efficiency, and believability.

Mohseni et al. [62] conducted comprehensive research into different XAI techniques, culminating in a proposed design and evaluation framework which takes into consideration the design goals and evaluation measures for different target user groups. This framework iteratively cycles between design and evaluation "poles," traversing inwards starting with overall XAI system goals, moving to the end-user explainable interface, and finishing with the underlying interpretable algorithms. This framework acts as a unified model, guiding appropriate evaluation measures and methods for each layer according to the aforementioned target users and their design goals. For AI novices these methods are aligned with what is proposed by [40], evaluating the user's mental models, levels of usefulness, satisfaction, trust and reliance, and measuring human-AI performance. Data experts wish to evaluate task and model performance, whilst AI experts prioritise explainer fidelity and model trustworthiness.

Das and Rad [17] discuss the relatively new area of evaluation techniques for explanations, which has primarily focused on human-in-the-loop (qualitative) explanations. Research from [17] identify key constraints for effective, human-useable explanations: *identity* (identical instances must produce identical explanations), *stability* (same class of instances must produce comparable explanations), *consistency* (instances with a singular feature changed must produce explanations which amplifies this change), *separability* (instances from different populations must produce explanations with notable dissimilarity), *similarity* (regardless of class differences, close instances should produce similar explanations), *implementation constraints* (methods should minimise time and computational complexity), and *bias detection* (testing dataset should have detectable bias).

Furthermore, the authors [17] provide current schemes for evaluation techniques:

- System Causability Scale (SCS) [41]: A measurement tool designed to evaluate the quality of explanations produced by XAI. Specifically, how well an explanation enables a human expert to understand the *cause* of a result.
- Benchmarking Attribution Methods (BAM) [88]: A framework for systematically evaluating and comparing feature attribution methods used in XAI.
- Faithfulness and Monotonicity: This measures the correlation between importance scores of features and the performance effect of each feature towards a correct prediction.
- Human-grounded Evaluation Benchmark [61]: A method to understand the precision of XAI explanations compared to human generated explanations.

In order to quantitatively evaluate LLMs as conversational explanatory agents, a composite ACF score was produced. This score draws from established methods in research for evaluating XAI techniques. In particular, it uses the Co-12 properties of correctness, completeness, and coherence. Furthermore, the use of reference human answers aligns with the human-grounded evaluation benchmark.

2.3 Explainable AI Question Bank (XAI-QB)

Although explanations have been found to improve user understanding, conclusions about user trust and acceptance remain mixed. This acceptance is dependent on the user's own mental models, use-case, and domain knowledge —for which the majority will lie outside of the scope of AI research, and therefore outside the technicalities of the XAI features attempting to explain them. Fundamentally, explanations are a psychological construct —answers to a *why?* question driven by the users personal desire to understand [59, 40]. The effectiveness and overall adoption of explanations are relative to the user, therefore posing an inherent challenge to bring together a diverse set of technical capabilities and explanation features.

To address end-user explainability needs, and enhance the growing collection of XAI tools, the XAI Question Bank (XAI-QB) proposed by Liao et al. [48] is an algorithm informed set of prototypical questions a user may ask about an AI system. The XAI-QB is built on top of a taxonomy of current XAI methods used for: *explaining the model (global), explaining a prediction (local), inspecting a counterfactual*, and *providing examples*. Within these methods exist the primary questions: *How? Why? Why not? What if? How to still be this?* In addition, the dimensions: *data, output*, and *performance* are included, which form what is referred to as the 9 explainability needs categories. The XAI-QB aims to inform best practices for AI system development and design, and provides heuristic guidance to ensure each of these explainability needs categories have been satisfied.

Furthermore, the XAI-QB can be deployed in user research to more appropriately specify their needs. AI system developers and designers may struggle to alleviate technical and practical barriers between XAI methods and their users. The XAI-QB ensures that XAI methods direct their attention towards techniques that address user needs, moving towards a more human-centred approach which bridges the gap between between algorithmic output and user accepted answers. Liao et al. identified two properties of human explanations which offer guidance on how XAI tools can explain both naturally and effectively. Firstly, explanations are *selected*, focusing on only a few specific causes, rather than an exhaustive list, providing enough information to invoke trust without being overwhelming. Secondly, explanations are *social*, as part of an evolving conversation based on the user's current beliefs and how their own understanding changes over the course of the interaction. These two elements highlight the need for XAI to be interactive and conversational, tailoring explanations specifically to the user, with the ability to ask follow-up questions and iteratively fill gaps in their own knowledge. Spanning across the explanation needs categories:

Input/data allows users to appropriately evaluate the AI system's capabilities and limitations. This is especially important when the user is directly involved with, or has the ability to alter the data, and therefore influence performance and fine-tune an AI system (i.e. feedback loop). Relevant XAI methods allow the user to have transparency over the training data, investigating these limitations (e.g. biased, or missing data) and identify any regulatory concerns (e.g. transferability).

Similarly, the *output* also allows users to appropriately evaluate and better utilise the AI system's capabilities. In practice, motivations for explainability are driven by a user's system goals, such as decision-making and optimising processes. Explainability in the context of a specific use-case supports downstream tasks and enhances overall workflow. User demand indicates the need for XAI methods to explain how to best take advantage of the output and make actionable insights concurrent with the user's product goals.

Performance XAI methods are considered less critical in these decision contexts, especially for those who work outside of technical scope of AI and analytics. Users are not satisfied when given solely a recommendation or score such as accuracy, which may be hard to interpret and therefore make insights difficult. Furthermore, discrepancies between performance on training vs actual data may further increase user hesitancy, hindering both trust and understanding. Similar to the *input/data* category, performance is mostly used to investigate the limitations of an AI system (e.g. constraints on when/where it can and can't be used).

Model behaviour can be explained on both globally and locally, depending on the user's question and desired level of granularity. When explaining global model behaviour, the *how* category is used mainly by those with an AI or analytics background to improve their own mental models about the AI system,

and how it can be best utilised. The most demanded explanations are sought out on a local level after an unexpected or anomalous event, prompting the user to ask *why, why not?* XAI methods within this explanation category enable users to investigate individual predictions which are contrary to their own expectations and beliefs. Furthermore, *what if, how to be* questions promote interactive explanations, where users can explicitly reference an outcome and ask follow up questions. Inspecting counterfactuals is a less commonly used feature of XAI methods, but allow the user to further understand the boundary of system capabilities by testing different scenarios and seeing the effect on model responses.
Explaining *input/data* and output are considered "static" explanations which come up in the early stage of system usage. This is when initial assessments are made surrounding AI capabilities and limitations. Other explanation categories are regarded as "dynamic," which require more user involvement, and will be engaged much more frequently.
Supplementary to algorithm informed categories, *other* questions in the XAI-QB explore additional areas of interest for AI understanding. These include follow up questions as to why a certain feature or dataset is used, how the AI system adapts and influenced, and terminological questions. These types of questions naturally arise in an explanation interaction, improving the user's mental model about the AI system, its capabilities, limitations, and how it works.

2.3.1 Extending the XAI Question Bank

The XAI-QB is an influential piece of work in the field of XAI, largely referenced in literature review. However, the actual deployment of the XAI-QB in practical use-cases remain limited. Few studies actually test the validity of its questions in user studies, or reflect on its applicability. Sipos and Schäfer [79] conducted a qualitative study using the XAI-QB as a tool for end-user explainability, aiming to empirically evaluate the XAI-QB's suitability in practice. Sipos and Schäfer highlighted four key areas of their research:

- How to utilise the XAI-QB with end-users
- Proposing additional user-centred questions
- Refining current XAI-QB questions and examples
- Clarifying both the scope and applicability of the XAI-QB

In addition to amendments for all 49 questions included in the XAI-QB, a total of 11 further questions and a new explanation needs category, *User-interface (UI)*, is recommended. This new category was prompted by the fact that AI systems are not necessarily self-explanatory, with difficulties assessing the affordances of the system (i.e. what actions can be achieved), and particular UI design choices/functionalities (e.g. interactive components such as buttons). The additional questions are largely focused on the *output* category, with users demanding further insight into the source, scope and amount of data provided. Furthermore, users wanted to be made aware of the system's ownership and if it uses their interactions in any capacity (i.e. to change/adapt/improve) —indicating a desired level of responsibility and accountability. Although Sipos and Schäfer recognise the XAI-QB as a useful tool for exploring user explainability needs, they also highlight the need for further refinement. In practice, the current set of XAI-QB questions lacks coverage in certain areas (e.g. UI), with ambiguous terminology and certain questions difficult to differentiate. Consequently, this makes it especially challenging for a lay-user to utilise these questions, where the motivations for explainability relate more towards practically useful information, as opposed to the the inner-workings and technical specifications of the AI system. In its application, team discussions are required to properly define context-specific questions, align on terminology and definitions, and "customise" the XAI-QB for their use-case.

2.3.2 Nested Model for Design and Validation

Extending the XAI-QB even further, Dubey et al. [23] created a nested model for AI design and validation, aiming to facilitate AI regulation and ethical AI development. This expands the coverage of the XAI-QB to consider aspects of AI governance in both a rigorous and systematic way. The nested model traverses 5 distinct layers, spanning across the areas of: regulation, domain, data, modelling, and predictions. Each layer contains a set of prototypical questions which must be satisfied before progressing to the next. This is referred to by Dubey et al. as "*eliminating the potential threats of one layer through the functionality of the next layer.*" Using the XAI-QB as a foundation, these questions have been mapped to their relevant layers, and additional questions have been proposed where the XAI-QB lacks scope (such as in regulation or domain). Furthermore, this process can be iterative, with questions answered in one layer revealing additional questions in prior layers which need to be addressed. Different stakeholders have been identified as the key individuals tasked with answering questions within a certain layer, ensuring interdisciplinary collaboration between domain experts, machine learning practitioners, and end-users. This is also true for wider AI regulation efforts, which requires the cooperation between government, industry and civil bodies. AI regulations can be categorised into either ethical or technical requirements, determined by their focus area and objectives. Ethical regulations consider the objectives of fairness, transparency, accountability and privacy —seeking to preserve fundamental human rights and societal values. Whereas technical regulations are more concerned with details surrounding algorithms, data and implementing guardrails —ensuring secure, robust, and reliable systems. The nested model guarantees that both of these regulatory categories are acknowledged throughout the layers. The process of navigating the nested model provides a structured framework for AI system validation in both development and production, pairing regulatory science with AI.

Covering the entire scope of the AI system, this research project operationalises the extended XAI-QB as a schema for evaluating LLMs to answer these questions. Furthermore, the AI system was specifically created around these extended XAI-QB questions, informing design practices and ensuring HC-AI design principles are incorporated.

2.4 Large Language Models for XAI

Since late 2022 with OpenAI's release of the generative chatbot, ChatGPT, LLMs have received widespread acknowledgement and use, with their powerful ability to handle multimodal inputs and outputs. Trained on huge corpora of data, chatbots can draw from an expansive range of domains and technical expertise, allowing it to perform various tasks at an expert level. These LLMs, which have been fine-tuned to enhance their ability to follow instructions and display context-awareness, present a promising opportunity in the field of XAI, where complex processes can be simplified into human consumable text based narratives.

Bilal et al. [9] conducted extensive research into the current landscape of using LLMs for XAI, discussing the approaches and limitations to LLM explainability, techniques for evaluating such LLM generated explanations, and the ongoing challenges of achieving XAI using LLMs. Explainability as a whole can be segmented into three main approaches:

- *Post-hoc* methods (such as SHAP and LIME values, integrated gradients, and partial dependence plots) seek to clarify decisions after a model has been trained, analysing the feature contributions that influence the output. These methods are able to explain model behaviour on either a local (i.e. individual predictions) or global (i.e. overall model behaviour) level.
- *Intrinsic interpretability* relates to the notion of building models that are inherently understandable by their design, without the additional requirement of explainability tools. These models, such as linear regression, decision trees, and rule-based models enable greater transparency, with clearer relationships between inputs and outputs. This is particularly important in high-risk domains, where critical decisions on data-subjects/individuals need to be readily available and justifiable.
- Finally, *human-centred* explanations focus on the end-uer and their ability to comprehend how the AI system functions, implementing methods to optimise the socio-technical interactions between humans and AI. This considers *who* the explanations are for, as well as *what* and *how* it is being explained. Human-centredness takes into account user preferences, comprehension levels, and contextual factors to ensure explanations are the most effective and relevant for the audience and their use-case.

Evaluating across these explanation methods can be done in two key ways:

- *Qualitative* analysis investigates elements of comprehensibility and human understanding. This is the clarity a user has when provided an explanation, as well as the level of control through interactivity and adjustability (e.g. providing feedback, or asking follow-up questions). However, even if an explanation successfully provides information that is intelligible to the user, it may not be accurate to the model's actual logic, or the ground-truth.
- *Quantitative* analysis looks into these concerns surrounding faithfulness and plausibility, measuring the degree of accuracy an explanation has towards the technicalities of an AI system. This includes correctly representing aspects such as system design, internal modelling logic, data (input and output), causal relationships, and reproducible decisions. Furthermore, explanations need to align with domain knowledge, follow a logical consistency, and verify any cause-and-effect relationships.

Researchers have special benchmark datasets to improve their explanation generating capabilities when using LLMs for XAI. These datasets are designed to focus on specific aspects of explanations, such as reasoning, scientific knowledge, and ethical decision-making. [11, 69, 2] This allows researchers to train LLMs to explain model decisions by providing structured sentences which are clear and easy to understand. Despite LLMs being trained on a huge amount of data with billions of parameters, Bilal et al. highlight persistent challenges which LLMs are still facing. Certain considerations need to be made regarding the data used for training, or any additional data used when prompted for an explanation. Sensitive data needs to be handled in accordance with legal and regulatory standards. Data from disreputable or irrelevant sources may disproportionately influence weights/parameters and cause the LLM to stray away from faithful and plausible results. Societal and cultural differences also need to be accounted for, with robust data transferability required. This must be achieved at the same time as being cautious to not reinforce any social (e.g. age, gender, lifestyle), language (inaccessibility) or representation (skewed data) biases, which further entrench stereotypes and discrimination. LLMs face the ongoing challenge of meeting all levels of user demands and proficiencies. To greater achieve this, Bilal et al. state the need for future work using LLMs for the purpose of XAI to include cross-disciplinary collaboration between AI researchers, cognitive scientists, domain experts, and end-users. This ensures the full spectrum of stakeholders are incorporated into the process of tailoring LLM generated explanations. Furthermore, automating human feedback and tracking user interactions can be incorporated to constantly update and revise the explanations being presented. A possible choice being text based explanations generated by LLMs coupled with suitable visualisation techniques.

Similarly, Busra et al. [75] conducted a systematic literature review exploring research which utilises LLMs as both "explanation generators" and "XAI output translators." Current literature focuses heavily on quantitatively explaining local predictions to non-technical groups (e.g. SHAP and LIME values). This indicates that current methods prioritise the need for general audiences to be able to understand individual instances of model behaviour in a measurable way. LLMs aid in converting these technical XAI outputs into practical, context-aware descriptions.

Taking advantage of LLMs in-context learning (ICL) capabilities, Kroeger and Ley [45] showcased the use of LLMs as effective post-hoc explainers using two main stategies:

- *Perturbation-based ICL*: using LLMs to analyse a sample of local input variations to identify important features.
- *Explanation-based ICL*: using a sample of post-hoc explanations as ground-truth labels for LLMs generating new explanations.

These methods highlighted the in-context learning abilities of LLMs for generating faithful explanations on the same level as current leading post hoc methods (e.g. SHAP, LIME).

Slack et al. [80] created an interactive dialogue system, *TalkToModel*. Based on the user-specified model and data, XAI methods have been mapped onto different operations presented as SQL-like representations. User input (text) is passed through a *dialogue* engine which uses an LLM to parse it into a logical form of these operations, referred to as grammar. Validated by their ability to answer XAI-QB questions, this grammar provides a set of production rules (operations, arguments, relationships between operations) that an *execution* engine performs. The results of the *execution* engine are then used to populate operation specific text-templates. As opposed to using LLM generated text, templates are used to ensure stability, trustworthiness and avoid any verbosity or hallucinations by the system. This is then passed into an open-ended *text interface*, allowing the user to ask additional or follow-up questions. In comparison to a traditional "point-and-click" explanation interface (i.e. *explainerdashboard*), both domain and AI specialists preferred *TalkToModel* across metrics of easiness, confidence, speed, and likeliness to use. Furthermore, they displayed an ability to answer explanation-based questions faster and more accurately. Susnjak [81] proposed an LLM framework for augmenting predictive modelling with prescriptive, natural language feedback in higher education. This targets students deemed "at-risk" with proactive remedial attention, using counterfactual modelling (SHAP and DiCE [63]) to uncover a set of minimal, realistic changes for the desired outcome to be achieved (e.g. educational targets). These results are communicated to students and tutors via an LLM, which has been prompted to respond in a very specific, structured manner using example templates (employing explanation-based ICL).

Ali and Kostakos [3] developed *HuntGPT*, a cybersecurity system with an interactive UI to display intrusion detection results from a Random Forest (RF) classifier. Using ChatGPT 3.5 Turbo, the LLM displayed substantial proficiency in cybersecurity knowledge and its ability to communicate this information in an accessible and readable manner. Combined with XAI results (such as SHAP and LIME values), this interface facilitates the intersection of domain knowledge with the technicalities of AI, allowing threat-responders to make fast and informed decisions in critical moments.

Wellawatte and Schwaller [86] similarly created a UI interface, *XpertAI*, for understanding molecular structure-property relationships in chemistry. Wellawatte and Schwaller used an XGBoost model to predict the labels for molecular structures and extract the most impactful features (using mean SHAP values and LIME Z-scores). Using ChatGPT with chain-of-thought prompt-engineering, they allow the user to additionally include relevant scientific literature (either manually uploaded or scraped from arxiv.org), which results in cited, natural language explanations combining scientific findings/evidence with inferences from XAI output.

Tursun et al. [82] created a framework for making heatmap-based XAI automatic, scalable, interpretable and interactive. Similar to Slack et al. [80], this process involves two modules: a *context* module extracts task specific information from the input image (object recognition and localisation) and its corresponding heatmap (salient regions and dominant colour regions). This is formatted into a structured text-based caption and coupled with a specific question. This question serves as the initial prompt for the *reasoning* module, in this case ChatGPT 3.5, which creates an XAI report and opens up a dialogue for future questions and clarifying comments.

Also utilising ChatGPT 3.5 Turbo, Metzger et al. [56] created *Chat4XAI*, which provides natural language explanations about a deep reinforcement learning (RL) model's decisions. Leveraging prompt engineering, contextual information, and the RL specific XAI method "Decomposed Interestingness Elements" (DINEs), Metzger et al. were able to achieve results with high fidelity and stability, outperforming a user study of software engineers using only DINE visualisations to describe the model's decisions.

Bhattacharjee et al. [7], experiment with the instruction following capabilities and textual understanding of LLMs when black-box text classifiers are applied to NLP text classification datasets. Intending to facilitate causal explainability via counterfactual explanation, their framework identifies latent/unobserved features in the input text. From these the latent features they identify the associated input features (i.e. words), and can then use these input features (i.e. words) to generate a counterfactual explanations that try to preserve high semantic similarity with the original input.

The research from Dhaini et al. [20], integrates an LLM interface to strictly describe the outputs from explanation methods such as SHAP, LIME, and integrated gradients. Thus, supporting decision-making and enhancing the comprehensibility of these post-hoc methods, especially to lay-users.

The use of LLMs for XAI has previously either been restricted to templated outputs, which limits their conversational capacity, or evaluated solely by qualitative user studies measuring satisfaction and acceptance. This research project utilises LLMs as conversational explanatory agents and moves towards rigorous quantitative evaluations. This is achieved by the composite ACF score, constructed from the components of accuracy, completeness and faithfulness, alongside readability benchmarking. This aims to provide the necessary validation of LLMs as conversational explanatory agents before their deployment into high-risk settings.

2.5 Evaluating Large Language Model Explanations

Liang et al. [47] proposes the Holistic Evaluation of Language Models (HELM), which aims to go beyond traditional benchmarks when evaluating language models, and provide a more complete understanding of their capabilities and limitations. HELM firstly creates the design space for language model evaluation, consisting of a taxonomy of scenarios and metrics. The scenarios are a tuple of *task* (e.g. question answering, summarisation, information retrieval), *domain* (e.g. news, books), and *language* (e.g. English varieties), with domain further sub-divided into *what* (text e.g. books, news, social media), *who* (speaker e.g web users, gender, race, age), and *when* (time/circumstances e.g. 2025, pre-Internet). Reflecting a range of societal considerations, evaluation metrics cover the language model's accuracy, calibration, robustness, fairness, bias, toxicity, and efficiency. Similar to the research from Nauta et al. [64], the authors take a multi-faceted outlook in their evaluation metrics, recognising trade-offs (e.g. strong (inverse) correlation between accuracy, and robustness and fairness) and optimising for a desired objective.

Datta and Dickerson [18] advocate for a human-centred approach when evaluating the output of LLMs. They argue there is considerable transferability between the well-founded field of XAI research and emerging space of LLMs, both of which assist in supporting some downstream task. Both XAI and LLMs influence decision-making, with users continually revising their mental models when deciding to use, trust, and make sense of the outputs. This is largely influenced by their understanding of how the system works and their ongoing interactions with it. As a result, cognitive engagement needs to be constantly assessed, to ensure users are adequately identifying incorrect or misguided (and in the case of LLMs, hallucinated) outputs, especially as these systems are being deployed to assist in increasingly high-risk domains and important tasks. Furthermore, this downstream performance in real world scenarios (e.g. coding/debugging) needs to be measured to validate the utility of these tools as actual enhancements in accuracy, efficiency, or some other defined objective. This offloading of cognitive labour must not come at the cost of deskilling or replacing human abilities.

Gu et al. [37] created Digital Socrates to automatically evaluate (both qualitatively and quantitatively) the intrinsic quality of LLM reasoning chains. Given the question, explanation, and answer, the authors introduce an *explanation critiquing* task, which systematically evaluates explanation quality. Explanation flaws, if any, are localised, described, and categorised. These categories are chosen to cover different types of Socratic questioning: *misunderstanding*, *lack of justification*, *incorrect/missing information*, *incorrect/incomplete reasoning*, *inconsistent answer*, or *irrelevant*. These rigorously evaluate if proper understanding and reasoning has been exhibited by considering appropriate perspectives and logical implications. General and specific recommendations for correcting these flaws are then provided, to improve both broad model failures and offer guidance towards the right reasoning chain/answer. Finally, the explanations themselves are given a quality score from 0 (completely wrong) to 5 (completely correct). This culminated in an open source model and critique bank dataset containing reasoning-specific questions, answers, explanations and their corresponding explanation critiques.

Dhaini et al. [20] is a framework for benchmarking post-hoc feature attribution methods applied to transformer-based NLP models (e.g. BERT, GPT). Supported explainers can be either *gradient* (i.e. Saliency, Gradient x Input, Integrated Gradients, DeepLift, Guided BackProp) or *perturbation* (i.e. LIME, SHAP, SHAPIQ) based methods. Allowing for the comparison across these methods, the explanations are further assessed on their qualities of *faithfulness* (i.e soft comprehensiveness, soft sufficiency, feature attribution dropping curve, normalised area under curve, area under threshold performance curve), *plausibility* (i.e. intersection over union f1 score, token level f1 score, area under precision recall curve), and *complexity* (i.e. complexity, sparseness). This provides users with a diverse set of explanations, enabling comprehensive evaluation tailored to their use-case, and ensuring a balanced assessment of model fidelity (i.e. how well explanations align with model reasoning), human interpretability, and explanation brevity.

Ichmoukhamedov et al. [43] uses LLMs two-fold, as “explanation generators” and “feature extractors.” These are used to convert tabular SHAP data into a textual narrative, and then quantitatively evaluate the explanation with the SHAP values serving as the ground-truth. The narrative generation LLM is instructed to follow particular content and formatting rules to avoid ambiguity. From this narrative, the feature extraction LLM identifies the rank (i.e. relative importance of the feature), sign (negative or positive contribution to the SHAP score), value (i.e. actual value of the feature) and assumption (i.e. single sentence, if exists). From this, the authors quantitatively measure the elements of *faithfulness* (i.e. how accurately the narrative reflects the SHAP values and thus the predictive model (assuming SHAP values are also faithful)), *human similarity* (i.e. narrative compared against reference text written by human experts), and *assumptions/plausibility* (i.e. narratives containing additional information which leverage the LLMs expansive knowledge base). Rank, sign and value are measured against ground-truth SHAP values to asses faithfulness. Assumptions are measured through “perplexity,” which calculates how unlikely it is for the LLM to produce a given string. This is used as a rough proxy for the purposes of fact checking and capturing potential hallucinations. Finally, cosine similarity metric is used to find the distance between the LLM generated narrative and human expert written text, measuring the semantic similarity.

Dalal et al. [15] proposes *Inference to the Best Explanation (IBE)* which estimates the plausibility of LLM generated explanations through evaluations of logical consistency, parsimony, coherence, and linguistic uncertainty. This framework then selects the most plausible LLM generated causal explanation, shown to be in line (highly correlated) with human judgement.

Min et al. [60] developed FActScore (Factual precision in Atomicity Score), an evaluation metric used to measure the factual accuracy of text generated by LLMs, verified by a given knowledge source. An LLM generated response is broken down into smaller individual statements called "atomic facts," which contain a single piece of information. Each of these facts are checked against a trusted knowledge source (e.g. Wikipedia) to confirm if the statement is true or not. The result is a percentage score of atomic facts which are verifiably true. This was designed to combat hallucinations and extend beyond BLUE and ROUGE metrics which simply measure text similarity, but not factual correctness.

This research project takes inspiration from a selection of these methods for evaluating LLM generated responses. Central to this is the notion of *faithfulness*, where the logical consistency and semantic meaning is measured. This can be achieved with human-grounded evaluations (i.e. reference human answers). Furthermore, "atomising" responses into single pieces of verifiable information has been incorporated to measure the completeness of an LLM generated response.

3 Materials

3.1 Data

Flu Shot Learning: Predict H1N1 and Seasonal Flu Vaccines

DrivenData is a data science competition website focused on social impact, providing a range of datasets spanning across high-risk domains such as public health, education, conservation, and international development. These competitions connect organisations such as NGOs, research institutions, and governmental agencies with data scientists and machine learning experts to tackle pressing social challenges. The dataset selected comes from DrivenData's "Flu Shot Learning: Predict H1N1 and Seasonal Flu Vaccines" competition [22], and is originally sourced from the National 2009 H1N1 Flu Survey [12].

This dataset is provided in two csv files, with one file containing a selection of protected (e.g. sex, age, education, employment), behavioural (e.g. social distancing, mask wearing, hand washing), opinion (H1N1 concern, doctor vaccine recommendation), health (e.g. chronic medial conditions, health insurance) and socio-demographic (e.g. geographical region, employment industry) features, used for predicting the uptake of the H1N1 and seasonal flu vaccine. The other file contains the corresponding labels for the observations. This is a binary classification task where the respondent is self reporting whether or not they have received the vaccine (i.e. Yes = 1, No = 0).

3.2 Resources

A selection of packages and resources which are key to the implementation of this project, beyond standard practice, which are discussed here further.

3.2.1 Communicating with LLMs

ChatGPT, Claude, Gemini

Flagship models from OpenAI ($\text{ChatGPT}_{5.2}$) [67], and Anthropic ($\text{Claude}_{\text{Opus 4.6}}$) [4], and Google ($\text{Gemini}_{\text{3 Pro}}$) [35] were utilised for prompt engineering, response generation, and integration through their respective APIs. Source code of the AI system was uploaded directly to their native platforms for iterative prompting and manual evaluation to be conducted.

GitHub Marketplace Models

GitHub is a cloud-based platform where you can work independently or collaborate with other users on code. Through their marketplace, they provide free access to generative AI models directly within GitHub, allowing access to some of the most popular LLMs. These models are accessed through GitHub's API.

Azure AI Core and Inference

In combination with GitHub's API, Microsoft's Azure AI core and inference packages were used to create a client application. These large language models from GitHub's marketplace were accessed via Azure AI's ChatCompletionsClient (`azure.ai.inference.ChatCompletionsClient`) and AzureKeyCredential (`azure.core.credentials.AzureKeyCredential`) classes.

LangChain

LangChain is an open-source framework for building applications powered by LLMs, designed to make it easier for developers to integrate LLMs into complex applications that need context, memory, external data, or multi-step reasoning. This was the main architecture used to construct a controlled and reproducible conversational chatbot, connected via their respective API keys. This structure allows for parameters such as `temperature, top_p`, and `max_tokens` to be set.

3.2.2 Explainability Tools

The technical details are discussed further below in the section Explanations.

DALEX

moDel Agnostic Language for Exploration and eXplanation (DALEX) [8] is a suite of explanation tools which can be applied to a variety of different models. The primary function `explain` is a wrapper which creates an explanation object for a predictive model. This model can then be explored with a diversity of local and global explanation functions, such as (grouped) feature importance, ceteris paribus profiles, breakdown attributions, and Shapley values.

SHAP

SHapley Additive exPlanations (SHAP) [52] is a game theoretic approach to explain the output of any machine learning model. Using Shapley values [76] from game theory, it connects optimal credit allocation with local explanations. Similar to DALEX, an `Explainer` wrapper function computes SHAP values for individual predictions, calculating the contribution of features towards that prediction. This can be explored through bar, scatter, waterfall, force, and beeswarm plots.

3.2.3 LLM Response Evaluation

The technical details are discussed further below in the section Evaluation Methods.

Hugging Face

Hugging Face is a leading open-source AI platform and community that builds machine learning models, applications, and datasets. The `evaluate` package is used to instantiate the BERTScore evaluation tool [90], which leverages contextual embeddings from a selection of different BERT language models [19].

This automatic evaluation metric is used to measure the precision, recall, and F1 scores between our LLM generated response and human generated answers. Hugging Face also offers thousands of state-of-the-art pre-trained transformer models, made available with the `sentence-transformers` package. Embeddings for the LLM generated response and human generated answers utilised these models before computing their semantic similarity.

Azure AI Evaluation

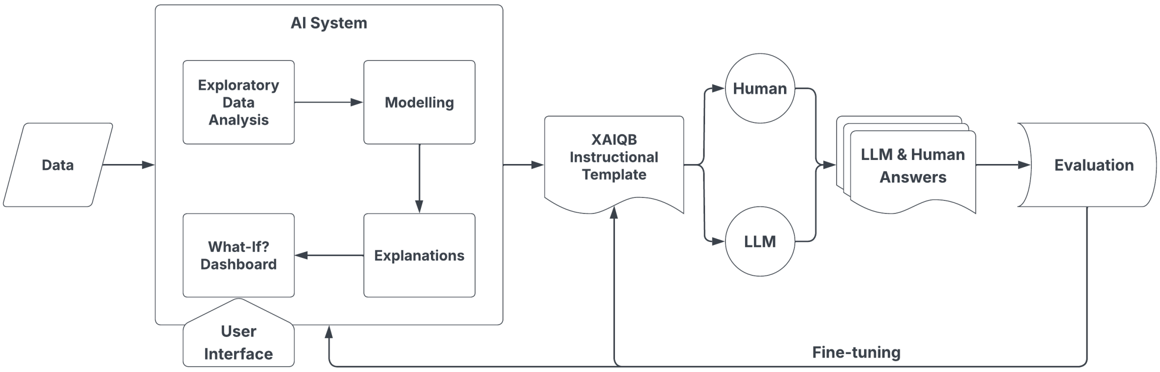
Azure AI Evaluation client library was used to assess the performance of our LLM generated responses in comparison to our reference human generated answers. In particular, this project utilised the `ROUGEScoreEvaluator` to compute ROUGE-L recall values [49].

Textstat, SpaCy

The `textstat` package was used to compute the readability metrics for both the human and LLM generated answers to the XAI-QB questions. These metrics are typically scored on a US grade level scale, allowing for greater interpretability of the scores and comparisons between evaluation metrics. The `spacy` package was used to compute summary statistics for the LLM generates responses, measuring their verbosity and number of difficult words.

4 Methodology and Study Design

4.1 Process Flowchart



4.2 Healthcare Domain

This AI system is constructed as a prediction tool and explanation interface for application in the healthcare domain, simulating a high-risk system utilised for AI-assisted decision-making. The specific task outlined is the binary classification of H1N1 and seasonal flu vaccine uptake (i.e. Yes = 1, No = 0), considering a selection of protected, behavioural, opinion, health, and socio-demographic features. This use-case is an important reflection of how an AI system can be used by clinical practitioners in a real-world scenario. The AI system can provide insights about population-level behaviour for public health vaccination campaigns and targeting strategies, or an individualised pre-consultation setting with personalised feedback and effective changes that mitigate vaccine hesitancy.

4.3 XAI-QB Selection

For this experiment, questions which inquired about local model behaviour were excluded, as these focus on individual instances/predictions which would require manual selection. This refers to all questions from the categories: *Why*, *Why not*, *How to be that*, *How to still be this*, and *What if*. A total of 36 questions from the extended eXplainable AI Question Bank were selected, focusing on global AI system behaviour and architecture.

4.4 Creating the AI System

In order to gauge the abilities of an LLM as a conversational explanation tool, an AI system was constructed for the LLM to analyse, evaluate, and explain. The LLM would have access to the complete source code of the AI system, which would then be asked the selected XAI-QB questions. The source code was developed with the intention of clearly providing the answers to all of these questions. These questions have been mapped to the sections of the source code where the answer can be located, with the full mappings provided in the Appendix.

The AI system is split across 4 sections:

1. Exploratory Data Analysis, **2. Modelling**, **3. Explanations**, **4. What-If? Dashboard**

Comprehensive documentation occurs within each section. Every snippet of code has been commented in order to describe what the code is doing and to explain the logic. Each section leads into the next with conclusions written more formally in markdown. Overall, exploratory data analysis performs feature selection for the modelling process. Several different models are trained and tested, with the highest performing model being selected for deployment. The architecture of this model is then explained, investigating the feature importance and influence (e.g. partial dependence plots, accumulated local effects). The most importance features are then selected to be present in the What-If? dashboard.

4.4.1 Exploratory Data Analysis

In this section of the AI system, the data was described, downloaded, and exploratory data analysis was performed. The data types were defined in advance, and a data dictionary was created which includes a definition of the features and any mappings of numerical values to their meaning (e.g. a Likert scale used to indicate how worried an individual is about the risk of the H1N1 virus, from 1 = "Not at all worried" to 5 = "Very worried"). An inspection and the dimensions of the dataset was extracted. Custom functions were also created to give clear descriptions of the numerical and categorical values in the dataset. For numerical values, the min, mean, max, standard deviation, [25, 50, 75]% quartiles and number of missing values were evaluated. For categorical values, the number of unique, most frequent, and missing values were evaluated.

In conclusion The target variables were analysed, inspecting the distribution of respondents who received the H1N1 vaccine, the seasonal flu vaccine, and both. It was decided to only model the H1N1 vaccine, and exclude seasonal features to avoid any feature leakage.

4.4.2 Modelling

Several different models were evaluated by the AI system. Data was split into training and testing using a standard 80:20 ratio, and a pipeline was created to process the data in preparation for modelling. Missing numerical values were imputed with the median (as all of the numerical data types are integer values, using the median preserves this), and categorical values were replaced with the string "missing." Furthermore, categorical values were one-hot encoded to convert them into a sparse binary matrix. Four models were evaluated to display a range of different model architectures and complexities that could be applied to this classification task.

Logistic Regression

This is a statistical model predicting the probability of a binary outcome. It applies a sigmoid function to the results of a linear regression model.

$$z = \beta_0 + \beta_1 x_1 + \ldots + \beta_n x_n, \qquad P(y = 1|X) = \sigma(z) = \frac{1}{1 + e^{-z}}$$

where $P(y = 1|X)$ is the probability the output is 1, given X, $\sigma(z)$ is the sigmoid function, and $\beta_0, \beta_1, \ldots, \beta_n$ are model coefficients. The model finds the best coefficients through maximum likelihood estimation (i.e. minimising the log-loss).

Random Forest

This is an ensemble machine learning method, aggregating the results from many different individual decision trees which have all been trained on different subsets of the data and its features (i.e. data bootstrapping and feature bagging). Training of decision trees is done independently and in parallel. In classification, the majority vote "wins."

$$\hat{y} = \text{mode}\{f_1(x), \ldots f_n(x)\}$$

where $f_i(x), i = \{1, \ldots, n\}$ represent n number of individual decision trees.

XGBoost

Similarly, this is also an ensemble machine learning method, but trains decision trees sequentially, learning from the errors of the previous trees. It begins with a naive prediction, computes the errors, fits a new decision tree to predict the errors, and updates the predictions (old prediction + η × tree prediction, where η is a learning rate). This is done until the model converges or the tree limit is reached. In boosting, this is the sum of weak learners.

$$\hat{y} = \sum_{i=1}^{n} f_i(x)$$

where f_i is the tree at iteration i, and n is the total number of individual decision trees. Each tree is trained to minimise a regularised loss function.

Multi-layer Perceptron (MLP)

This is an artificial neural network made up of many layers. The input layer receives the features, hidden layers identify and learn about patterns in the data, and the output later generates a final prediction. The hidden layers contain nodes which take the input, applies a weighted sum, passes the results through a non-linear activation function, and sends this output to the next layer.

$$z = w_1 x_1 + w_2 x_2 + \ldots + w_n x_n + b, \qquad a = f(z)$$

where x_i are the inputs, w_i are the weights, b is the bias, $f(\cdot)$ is an activation function (e.g. sigmoid, ReLu), and a is the output of the node. Training is done through backpropogation (forward pass, loss computation, backward pass, update weights), which adjusts the weights and biases to minimise a loss function, repeated many times (epochs) until the model learns the best weights.

Hyperparameter tuning

Hyperparameter tuning was performed for the Random Forest, XGBoost, and MLP models using a randomised 5-fold cross validation grid search, with a range of values provided for key parameters. 50 iterations were selected across the 5-folds, totalling 250 fits for each. The best parameters were selected based on scoring of the ROC-AUC value. The hyperparameter tuned XGBoost model scored the highest, and was therefore selected for deployment. This is a simplified model evaluation and selection process for the purpose of this research project, with more thorough validation required for application in the healthcare domain.

Predictions

The hyperparameter tuned XGBoost model was used to generate predicted probabilities for the positive class (i.e. the respondent receives the H1N1 vaccination, Yes = 1). These probabilities, p, were then binarised using a threshold where $\hat{y} = \mathbf{1}_{\{p \geq 0.5\}}$.

In conclusion The training and testing data, hyperparameter tuned XGBoost model, and predictions were exported as a pickle file to be investigated further with DALEX.

4.4.3 Explanations

Confusion matrix and classification report

A confusion matrix [44] is a 2x2 matrix which compares binary predictions with their labels (i.e. actual values). This allows for the evaluation of certain key metrics such as precision, recall and accuracy.

		Predicted Class		
		Positive	**Negative**	
Actual Class	**Positive**	True Positive (TP)	False Negative (FN) Type I Error	**Recall** $\frac{TP}{TP+FN}$
	Negative	False Positive (FP) Type II Error	True Negative (TN)	**Specificity** $\frac{TN}{TN+FN}$
		Precision $\frac{TP}{TP+FP}$	**Negative Predictive Value** $\frac{TN}{FN+TN}$	**Accuracy** $\frac{TP+TN}{(TP+TN+FP+FN}$

The classification report computes similar precision, recall and accuracy scores but split across each class in the model. Additionally, the harmonic mean of precision and recall is computed, called the F1 score:

$$F1 = 2 \cdot (\text{Precision} + \text{Recall})/(\text{Precision} \cdot \text{Recall})$$

The support value reports the number of actual occurrences of each class. Finally, mirco, macro, and weighted averages are computed for precision, recall, and F1 scores. This helps identify imbalanced datasets and performance of the model across each class.

AOC-ROC

Area Under the Receiver Operating Characteristic curve (AUC-ROC) [38] evaluates how well a binary classification model can distinguish between the classes. The model it's evaluating outputs a predicted probability on the interval [0, 1]. This probability can then be binarised based on a selected threshold value (i.e. $\hat{y} = \mathbf{1}_{p \geq \text{threshold}}$). This curve visualises how performance of classification changes as you move this threshold from 0 to 1, plotting the true positive rate (i.e. recall) $\frac{TP}{TP+TN}$ against the false positive rate $\frac{FP}{FP+TN}$. A perfect classifier has an AUC score of 1.0, with 0.5 being a random guess.

Feature importance

Feature importance [29] identifies which features are most influential for the model when making predictions. DALEX takes a model agnostic approach and evaluates feature importance via permutations, using the dropout loss. First, the loss function value (e.g. log loss for binary classification) is computed for the full model and its original data. The response variable is then shuffled and the loss is recomputed to indicate the worst possible loss function value, where there is no predictive signal in the data. This gives a baseline score to be used as a benchmark. Finally, for each explanatory variable, it is randomly shuffled across observations so it is not longer aligned with the response, and the loss is recomputed. An increase in the loss means a greater importance in the feature.

Partial Dependence Plots

A partial dependence plot (PDP) [32] shows the effect of a feature on the predicted outcome, visualising the relationship between an input feature and its target response. This is achieved by marginalising over the values of all other input features. In other words, the average effect on the prediction as we iterate over all possible values for the feature of interest, while keeping the remaining features the same as the original data.

$$\hat{f}_S(\mathbf{x}_S) = \mathbb{E}_{\mathbf{X}_C}[\hat{f}(\mathbf{x}_S, X_C)]$$

where $\mathbf{x}_S$ is the feature of interest, X_C are the remaining features, and $\mathbb{E}_{\mathbf{X}_C}$ is the average over all combinations of the other variables. $\hat{f}_S$ is estimated by calculating averages in the training data $\hat{f}_S(\mathbf{x}_S) = \frac{1}{n}\sum_{i=1}^{n} \hat{f}(\mathbf{x}_S, \mathbf{x}_X^{(i)})$. This tells us for a given value of the feature S, what the average marginal effect on the prediction is.

Accumulated Local Effects

Addressing limitations of partial dependence plots, accumulated local effects (ALE) [5] also shows on average how a feature influences a prediction, but accounts for the fact that features may be correlated with one another. This is addressed by calculating the differences in predictions, instead of using averages. The feature of interest is divided into a range of intervals (determined by the quantiles of the distribution, ensuring the same amount of data is contained in each of the intervals). For each interval, the model's average change in prediction is computed (i.e. the local effect). The local effects on an interval is then summed (i.e. accumulated).

$$\hat{\tilde{f}}_{j,ALE}(\mathbf{x}) = \underbrace{\sum_{k=1}^{k_j(\mathbf{x})} \underbrace{\frac{1}{n_j(k)} \sum_{i:x_j^{(i)} \in N_j(k)} \underbrace{\left[\hat{f}(z_{k,j}, \mathbf{x}_{-j}^{(i)}) - \hat{f}(z_{k-1,j}, \mathbf{x}_{-j}^{(i)})\right]}_{\text{Effect}}}_{\text{Local}}}_{\text{Accumulated}}$$

This is then centred so the mean effect is zero $\hat{f}_{j,ALE}(\mathbf{x}) = \hat{\tilde{f}}_{j,ALE}(\mathbf{x}) - \frac{1}{n}\sum_{i=1}^{n} \hat{\tilde{f}}_{j,ALE}(\mathbf{x}_j^{(i)})$

Shapley values and SHAP

SHapley Additive exPlanation (SHAP) [52] values is a method to explain individual predictions by quantifying the contribution of each feature to that prediction, derived from Shapley values in cooperative game theory. The Shapley value [76] of a feature represents its average marginal contribution to the overall prediction, computed over all possible combinations of feature subsets:

$$\phi_j(\text{val}) = \sum_{S \subseteq \{1,\ldots,p\} \setminus \{j\}} \frac{|S|!(p - |S| - 1)!}{p!}(\text{val}(S \cup \{j\}) - \text{val}(S))$$

where S is a subset of features used in the model, p is the number of features, val(S) is the prediction for feature values in set S that are marginalised over the features which are *not* included in S. All possible coalitions (sets) of feature values have to be evaluated with and without the j-th feature to calculate the exact Shapley value.

SHAP estimates these Shapley values using a permutation-based (Monte Carlo) approach, in which random subsets of features are sampled from permutations. For each random permutation, feature contributions are computed incrementally by adding features one at a time (forwards and backwards using antithetic sampling). This process is repeated for many random permutations, and the marginal contributions for each feature are averaged to obtain an estimate of its Shapley value:

$$\hat{\phi}_j^{(i)} = \frac{1}{2m}\sum_{k=1}^{m}(\hat{\Delta}_{o(k),j} + \hat{\Delta}_{-o(k),j})$$

where $\hat{\Delta}_{o(k),j}$ is the k-th marginal contribution of feature j and $-o(k)$ denotes the corresponding reverse permutation.

In conclusion After inspection of the model architecture, features were chosen based on their importances scores to be used within the What-If dashboard. The top three most important numerical features, and top three most important (non-anonymised) categorical features were selected. These are: Doctor recommends the H1N1 vaccine, opinion the H1N1 vaccine is effective, opinion the H1N1 virus is a risk, age group, race, and sex.

4.4.4 What-If? Dashboard

The What-If? Dashboard is an interactive prediction tool and explanation interface which incorporates UI design into the AI system. This dashboard allows the user to change the six most important features of a patient through the use of select boxes. The remaining numerical features have been imputed with their median value, and the remaining categorical features have been imputed with the most frequent value. This is in order to create an "average" baseline observation for which these changes can be made. Furthermore, a slider allows the user to adjust the prediction threshold based on how conservative they want their analysis to be. This sets the minimum predicted probability required to determine if the observation will receive the H1N1 vaccine (i.e. $\hat{y} = \mathbf{1}_{\geq \text{threshold}}$). Once the selections have been made, a new prediction is generated with accompanying explanatory visualisations, including a prediction breakdown, Shapley values, and ceteris paribus profiles [46] for both the numerical and categorical variables.

4.5 Instructional Template

The instructional template used for prompting the LLMs was provided in JSON format. This was in order to ensure the instructions were being followed, and the output was produced in both a structured and repeatable manner. There are 4 main sections of the instructional template.

Metadata Provides important information about the template, such as the title, author(s), description (i.e. intended use), and version control.

Persona In order to make the answers domain relevant, a persona was added to prime the LLM to use specific terminology and suitable language. This persona informed the LLM it was to act as an "Explainable AI specialist communicating with healthcare professionals about the safe, transparent, and ethical use of an AI system in clinical and public health settings." It also defines the target audience, clarifying the intend recipient of these answers, and the tone in which the answers should be constructed.

Instructions This section was split into two subsections. **Guidelines** direct the LLM on how to answer the questions. This was introduced to enforce rules on how to explain the answers in a clear and clinically relevant way, enhancing the defined persona. The target audience are healthcare professionals and medical practitioners. Therefore, answers must not include overly technical terminology, or excessive use of mathematical formulas and code. The answer to the question has to be relayed in a clear, and clinically relevant manner. **Output** directs the LLM on how to provide the answers. For each of the selected XAI-QB questions there are placeholder strings it must fill with its answers. The output should only be a JSON with the completed question-answer pairs.

Questions These are the selected questions to the XAI-QB, formatted in a JSON file for the LLM to populate with its answers. This JSON is structured:

```
{"questions":
    {"category1":
        {"question1": {"answer": ""},
         "question2": {"answer": ""},
         ...
```

```
        },
    "category2": ...
    }
}
```

Reference human answers were produced in the same format as the **Questions** JSON.

The full instructional template can be found in the Appendix.

4.6 ChatGPT, Claude, Gemini

The source code of the AI system was uploaded directly to the native platforms of $\text{ChatGPT}_{5.2}$, $\text{Claude}_{\text{Opus 4.6}}$, and $\text{Gemini}_{\text{3 Pro}}$. These LLMs were also connected via their APIs for greater reproducibility with `temperature`, `top_p`, and `max_tokens` parameters. However, API access for these LLMs proved too restrictive for the full scope of this research project (i.e. `max_tokens` limits, context window, message trimming, file truncation), so the native platforms were utilised for the entire process. This will be further discussed in limitations and future work. The instruction template was fine-tuned iteratively through a process of prompt engineering —refining the inputs given to an LLM in order to reliably elicit more accurate, structured outputs — [74, 87], where the outputs were evaluated both qualitatively (empirical assessment) and quantitatively (evaluation metrics discussed below). The instructional template went through three major iterations, where the persona, instructions and questions were redesigned. This was mainly due to improper response formatting, clarifying the scope and intention of the template, and audience-centric (i.e. healthcare domain) persona development. In total, for each LLM the experiment was conducted three times. The source code (i.e. eda, modelling, explanations, what-if) was provided, and the LLM was prompted with the instructional template in a new instance. This was performed to additionally measure the LLM's output stability and response consistency to the same input (considering the weights of the model, such as `temperature` could not be explicitly set). In total, 9 runs were generated with 324 answers to evaluate.

4.7 Evaluation Metrics

4.7.1 Accuracy, Completeness, Faithfulness

Explainability, through its ability to provide transparency and enhance understanding, ultimately serves to build trust and promote acceptance of the system it's presenting. To ensure the answers generated by the LLM are factually correct and provide valid explanations, they can be assessed on three key metrics.

- **Accuracy** is the alignment of the answers generated with the ground-truth, and how closely it reflects this.
- **Completeness** considers if the answers capture all the key information without omissions.
- **Faithfulness** measures the consistency of the answers generated with the reference material. In this case, the human generated answers to the same questions.

These metrics were carefully selected to provide a composite score of quantitative XAI evaluation techniques, taking inspiration from current research literature. After a systemic review of over 300 XAI research papers, the Co-12 properties developed by Nauta et al. [64] outline *correctness* (i.e. accuracy), and *completeness*, as the most common XAI properties, with *coherence* (i.e. faithfulness) also being defined. In addition to these properties, supportive research literature (focusing specifically on LLM evaluation) for each individual metric is provided within their respective sections below.

Accuracy: Manual Evaluation

The correct answer to a question can be written in multiple different ways, and with varying degrees of accuracy. To evaluate the accuracy of the LLM generated responses, human generated answers for these XAI-QB questions were produced to serve as ground-truth labels and act as a marking scheme. However, manual evaluation introduces a degree of subjectivity, so a correctness scale was used to deliberately reduce bias. The answers were scored on an scale of:

$$0 = \text{Incorrect}, \qquad 0.5 = \text{Partially Correct}, \qquad 1 = \text{Fully Correct}$$

Incorrect answers are when the response fails to identify any relevant aspect of the AI system in relation to the question. This may be a generic response to the question without any reference to the source code and its output, or a complete misinterpretation of the question.

Partially correct answers are where some aspects of the AI system are used to answer the question, but does not cover all of the required topics. This means the answer is correct, but incomplete.

Fully correct answers cover all of the topics required to completely answer the question, referencing them correctly from the source code.

In the event that the answers provided are incomplete relative to the reference human answers, but extend the answer (correctly) in a different direction, this level of correctness is determined by the marker. This may occur when XAI-QB questions are more open-ended or allow for different interpretations. These are scored more intuitively than using the reference human answers as a definitive mark scheme. In these instances, manual evaluation can be used to identify which questions are vague and need to be clarified further, and detect any knowledge gaps within the human generated answers. Although manual evaluation is time consuming, this is a necessary element to HC-AI design principles, keeping humans in-the-loop and introducing a degree of human oversight.

The use of a correctness scale when evaluating LLM generated answers is precedented in research literature, with the authors from [37] producing a quality score from 0 (completely wrong) to 5 (completely correct). Furthermore, the authors [43] recognise that reference human answers are subjective and non-exhaustive, and therefore evaluate an LLM's responses for *assumptions/plausibility* (i.e. narratives containing additional information which leverage the LLMs expansive knowledge base).

Completeness: BERTScore and ROUGE-L

Information Units

In the task of evaluating the completeness of an LLM generated response, information units were constructed to identify the presence of essential terms. These information units were manually created and reduce the human generated answer into its core elements —containing key words, phrases, and figures which need to be included in order for an answer to be considered complete. This draws inspiration from FActScore [60], which atomises an LLM's response into individual, verifiable claims (i.e. "atomic facts"), which are then checked against a trusted knowledge base for its factual correctness. Instead, human reference answers were atomised into information units and checked within the LLM generated responses. This is done in order to create a single collection of information units (i.e. do not need to create new information units for every LLM generated response), and to have greater control (i.e. human oversight) over what is defined in the information units for a complete answer. These information units were provided as a list, which would be iterated over and checked against the LLM generated response.

Example 1:
Question: "What is the source of the training data?"
Human Generated Answer: "Data is sourced from DrivenData's 'Flu Shot Learning: Predict H1N1 and Seasonal Flu Vaccines' competition and originates from the US National 2009 H1N1 Flu Survey (NHFS). This survey was conducted between October 2009 and June 2010. It was a one-time telephone survey of households to monitor vaccinations during the 2009-2010 flu season in response to the 2009 H1N1 pandemic."
Information Units: ["DrivenData", "US National 2009 H1N1 Flu Survey (NHFS)"]

The term "source" can be interpreted as the origin of something, or where it was obtained. Therefore, we select the information units as "DrivenData" (i.e. where the data was downloaded), and "US National 2009 H1N1 Flu Survey (NHFS)" (i.e. where the data originates).

Example 2:
Question: "What is the sample size of the training data?"
Human Generated Answer: "There are a total of 26,707 observations with 37 features. This was split into training and testing data using an 80:20 split. Training data is 21,365 rows, testing data is 5,342 rows."
Information Units: ["21,365", "37"]

Sample size refers to the dimensions of a dataset. Therefore, the information units as "21,365" (rows) and "37" (columns) were selected. We only include the numerical values to avoid differences with how these dimensions are referenced (e.g rows can also be referred to as observations, individuals, records), and should not be penalised when evaluating the faithfulness of the response.

BERT

BERT (Bidirectional Encoder Representations from Transformers) [19] is a deep learning model developed by Google in 2018, which helps automate the understanding of words in context. This is achieved by reading the sentence in both directions (i.e. bidirectionally), providing a deeper contextual understanding of how words in a sentence relate to one another. Through the self attention mechanisms within its transformer architecture, weights are applied for the importance of different words in the sentence. BERT is pre-trained on large text corpora through masked language modelling (MLM) where words are randomly masked in a sentence and the model is trained to predict them, and next sentence prediction (NSP) where the model is taught the relationship between pairs of sentences.
This has been extended further with models such as RoBERTa (Robustly Optimized BERT Pretraining Approach) [51], which maintains the architecture of BERT, but refines the pre-training process through increasing the volume of data, larger batches, dynamic masking, and removing NSP.

BERTScore

BERT can handle synonyms, paraphrases and distant dependencies. By employing this language understanding, BERTScore [90] measures the semantic similarity of two sentences by computing the cosine distance between their contextual embeddings.
Given a tokenised reference sentence (i.e. human generated response), $x = \langle x_1, \ldots, x_k \rangle$ and tokenised candidate sentence (i.e. LLM generated response), $\hat{x} = \langle \hat{x}_1, \ldots, \hat{x}_l \rangle$, the embedding model (e.g. BERT, RoBERTa) generates these contextual embeddings as a sequence of vectors $\langle \mathbf{x_1}, \ldots, \mathbf{x_k} \rangle$ and $\langle \hat{\mathbf{x}}_\mathbf{1}, \ldots, \hat{\mathbf{x}}_\mathbf{l} \rangle$ respectively. The cosine similarity of a reference token x_i and a candidate token $\hat{x}_j$ is then given by:

$$\cos(\theta) = \frac{\mathbf{x}_i^\top \hat{\mathbf{x}}_j}{\|\mathbf{x}_i\| \|\hat{\mathbf{x}}_j\|}$$

Using pre-normalised vectors, this can be reduced to the inner product $\mathbf{x}_i^\top \hat{\mathbf{x}}_j$.
Greedy matching maximises the similarity score, where a token in one sentence is matched to the most similar token in the other sentence. Recall matches each token in x to a token in $\hat{x}$, while precision matches each token in $\hat{x}$ to a token in x. Therefore, recall was the selected metric, searching for token similarity of the information units within the LLM generated answer.

$$\mathbf{BERTScore}_{\text{Recall}} = \frac{1}{|x|} \sum_{x_i \in x} \max_{\hat{x} \in \hat{x}} \mathbf{x}_i^\top \hat{\mathbf{x}}_j,$$

BERTScore then undergoes optional importance weighting (where rare words can be more indicative of sentence similarity than common words), and baseline rescaling to increase score readability (placing BERTScore on an approximate interval of [0, 1]).

ROUGE-L

ROUGE (Recal-Oriented Understudy for Gisting Evaluation) [49] is a selection of metrics for evaluating machine translation and automatic summarisations in natural language processing.

ROUGE can be computed in numerous different ways. Whilst ROUGE-N metrics look at the overlap of n-gram exact word matches between two pieces of text, ROUGE-L finds the longest common subsequence, and is the metric of choice when measuring completeness. The longest common subsequence is calculated by taking two sequences (i.e. the LLM generated response, and the information unit to be identified), and finds the longest sequence that appears in both, in the same order, but not necessarily contiguously (i.e. a subsequence, which is derived from a given sequence by removing 0 or more elements, that do not have to neighbour each other, whilst preserving the order of the remaining elements).

ROUGE-$\mathrm{L_{Recall}}$ measures how much of the information unit appears in the LLM generated response. Therefore, it is the selected metric for evaluating completeness. This is a score on the interval [0, 1], with 0 indicating the information unit does not appear at all in the LLM generated response, and 1 indicates it appears fully.

$$\textbf{ROUGE-L}_{\text{Recall}} = \frac{\text{LCS}(x, \hat{x})}{\text{Unigrams in } x},$$

where x = Information Unit, $\hat{x}$ = LLM Generated Response

As validation, these information units need to have a ROUGE-$\mathrm{L_{Recall}} \geq 0.8$ when measuring against its reference human answer. Combining **ROUGE-L**$_{\text{Recall}}$ and **BERTScore**$_{\text{Recall}}$ provides the necessarily strict requirements for specific terminology and values to be included, with the flexibility of different phrasing and writing styles.

In addition to FActScore, this measure of completeness draws from research conducted by Gu et al. [37] when categorising LLM explanation flaws based on the Socratic principles: *incorrect/missing information* and *incorrect/incomplete reasoning*.

Faithfulness: SBERT

SBERT (Sentence-BERT) [70] starts with a pre-trained BERT model and adds a pooling layer (such as the mean of all output vectors) to derive a fixed length sentence embedding for each of the two sentences being compared. SBERT is then fine-tuned using pairs of sentences and their similarity labels from datasets such as the The Stanford Natural Language Inference (SNLI) Corpus [11] or Semantic Text Similarity (STS) dataset [13]. A Siamese network, which is an artificial neural network consisting of two identical subnetworks (same architecture, parameters, weights), independently processes two embeddings in parallel. During training, the weights are computed to minimise the distance of similar inputs, and maximise the distance of dissimilar inputs. Once trained, these sentence embeddings can be stored in advance, which ensures such sentence embeddings are semantically meaningful and can be compared with cosine similarity.

In literature, Ichmoukhamedov et al. [43] evaluates for *human similarity*, which also computes the cosine similarity between the LLM generated responses and reference text written by human experts. This faithfulness metric intends to capture the Socratic principles outlined by the authors [37]: *lack of justification*, *misunderstanding*, *inconsistent answer*, and *irrelevant*. Furthermore, research conducted in [15] estimates the plausibility of LLM generated explanations through evaluations of logical consistency and coherence, which is also intended to be captured by this faithfulness metric.

Both SBERT and BERTScore are essential metrics for evaluating the faithfulness and completeness of the LLM generated response to the human generated answer, allowing similarity to be evaluated at different levels of granularity. SBERT encodes full sentences into single vector representations, capturing the overall sentiment of the answers. In contrast, BERTScore operates on a token level similarity, matching individual words. This enables the evaluation of LLM generated response in terms of capturing the overall the sentiment of the answer, and the specific factual details included.

Accuracy - Completeness - Faithfulness (ACF)

The scores for accuracy, completeness, and faithfulness are computed individually. These metrics are then summed and normalised onto an interval of [0, 1] to give an overall *Accuracy - Completeness - Faithfulness* (ACF) score.

$$\text{ACF Score} = \frac{1}{3}\left[\underbrace{\text{Manual Scoring}}_{\text{Accuracy}} + \underbrace{\frac{1}{2}\left(\text{ROUGE-L}_{\text{Recall}} + \text{BERTScore}_{\text{Recall}}\right)}_{\text{Completeness}} + \underbrace{\text{SBERT}}_{\text{Faithfulness}}\right] \quad (*)$$

The parameters were chosen to give equal weighting to all three components within the composite ACF score. This provides a benchmark for further investigation and fine-tuning of these parameters, which can be changed based on different objectives. For instance, if an accurate answer is more important than semantic similarity to a reference human answer, or if completeness is a stricter requirement to ensure all topics are covered, beyond what's being provided is accurate.

4.7.2 Readability

Markus et al. [54], conducted a comprehensive survey about the role of explainability in creating trustworthy AI for healthcare. The authors determine an explanation should be understandable (i.e. readable) in addition to accurately describing the model behaviour. Readability is the assessment of how easily a reader can understand a written body of text. Having a higher readability level is crucial to ensure clear and accessible information, whereby the text can be more easily understood and connected to the reader's current knowledge base. This is especially important when disseminating complicated or technical information. Various scoring systems have been formulated to assess readability, with key components focusing on the number of words, sentences, and syllables. The formulas typically determine that longer sentences containing more words are structurally more complex, and therefore more difficult to read. Similarly, longer words containing more syllables are lexically more complex, and therefore more difficult to read. These scores are generally formulated to approximate the minimum grade level at which the text is considered to be understandable. These grade level scales are appropriate for use with domain experts, as they can be easily mapped to the level of education attained, from a 5th grade up to college graduates and academic professionals. This is an easily interpretable scale that can be used to quickly evaluate if the readability is adequate for the target audience, which is particularly important in this use-case, where healthcare professionals are being explained a blended view of technical AI/ML information with clinical evaluations.

Flesch-Kincaid Readability Test

The Flesch-Kincaid Grade Level [28] is an evolution of the Flesch Reading Ease Score (FRES) [30], which was one of the first data driven methods to try and quantify reading difficulty. These methods evaluate the structural and lexical complexity of a passage of text, but assign different weights to the length of sentences and words. The Flesch-Kincaid Grade Level was adapted in order to achieve a grade level equivalent Flesch Reading Ease Score, enabling easier comparisons and evaluations of text.

$$0.39\left(\frac{\text{total words}}{\text{total sentences}}\right)+11.8\left(\frac{\text{total syllables}}{\text{total words}}\right)-15.59$$

SMOG Index

SMOG stands for the "Simple Measure of Gobbledygook." [55] This metric aims to evaluate the number of years in education required to understand a body of text, and is commonly used for assessing the readability of health literacy materials.

$$1.043\sqrt{\text{number of pollysyllables}\times\frac{30}{\text{number of sentences}}}+3.1291$$

Coleman Liau Index

In contrast to the other metrics used to assess readability, the Coleman Liau Index [14] takes into account the number of characters as opposed to syllables, making this metric more consistent and easier to compute programmatically.

$$0.0588\cdot L-0.296\cdot S-15.8$$

where L is the number of letters per 100 words and S is the average number of sentences per 100 words.

Dale Chall Readability Score

Dale Chall Readability Score [16] takes into account the most commonly used 3000 English words, stipulating that passages of text are easier to read when the words are more familiar to the reader.

$$0.1579\left(\frac{\text{difficult words}}{\text{words}}\times 100\right)+0.0496\left(\frac{\text{words}}{\text{sentences}}\right)$$

5 Results

The ACF score is the composite metric of accuracy, completeness, and faithfulness. This is given by:

$$\text{ACF Score} = \frac{1}{3}\left[\underbrace{\text{Manual Scoring}}_{\text{Accuracy}} + \underbrace{\frac{1}{2}\left(\text{ROUGE-L}_{\text{Recall}} + \text{BERTScore}_{\text{Recall}}\right)}_{\text{Completeness}} + \underbrace{\text{SBERT}}_{\text{Faithfulness}}\right] \qquad (*)$$

For each LLM, the experiment was conducted three times. The source code (i.e. eda, modelling, explanations, what-if) was provided, and the LLM was prompted with the instructional template in a new instance. This is in order to to eliminate conversational memory or contextual carry-over between runs, ensuring that each evaluation reflects the LLM's response in isolation, rather than any accumulated context from prior interactions. This was additionally important to determine the variability of the LLM in response to the same information and prompt, when parameters such as `temperature, top_p`, and `max_tokens`, could not be explicitly set. In total, 9 runs were generated with 324 answers to evaluate.

Model	**Run**	**ROUGE-L$_{\text{Recall}}$**	**BERTScore$_{\text{Recall}}$**	**SBERT**	**Partial ACF Score**
ChatGPT$_{5.2}$	1	0.381	0.851	0.579	0.598
	2	**0.458**	**0.862**	**0.611**	**0.635**
	3	0.444	0.858	0.601	0.626
Claude$_{\text{Opus 4.6}}$	1	0.516	0.856	**0.593**	0.639
	2	0.466	0.852	0.585	0.622
	3	**0.530**	**0.857**	0.591	**0.642**
Gemini$_{\text{3 Pro}}$	**1**	**0.407**	0.865	**0.586**	**0.611**
	2	0.348	0.867	0.532	0.570
	3	0.399	**0.868**	0.579	0.606

The average range of scores across the LLMs for the component metrics are:

$$\text{ROUGE-L}_{\text{Recall}} = 0.067 \quad \text{BERTScore}_{\text{Recall}} = 0.006 \quad \text{SBERT} = 0.031 \quad \text{Partial ACF Score} = 0.033$$

These ranges are minimal across all LLMs and their component metrics. This highlights a sufficient level of output stability and response consistency in the LLM's responses, indicating a degree of reproducibility when using the LLMs native platforms for this analysis.

Manual scoring is time consuming. With the iterative process of prompting, evaluating, and fine tuning, this makes it infeasible beyond a quick empirical sense check. Therefore, for each LLM the run was selected based on the highest (partial) ACF score, computed from their completeness and faithfulness components: $\frac{1}{2}\left[\frac{1}{2}(\text{ROUGE-L} + \text{BERT}) + \text{SBERT}\right]$. This provides a clear indication about which run contains answers closest to the reference human answers, in terms of both the topics covered and their semantic meaning. As a result, run 2 was selected for ChatGPT, run 3 was selected for Claude, and run 1 was selected for Gemini to proceed to manual evaluation.

5.1 Accuracy, Completeness, Faithfulness

Accuracy

To evaluate the accuracy of the LLM generated responses, human generated answers for these XAI-QB questions were produced to serve as a ground-truth and marking scheme. The answers were then manually evaluated on an interval of:

$$0 = \text{Incorrect}, \qquad 0.5 = \text{Partially Correct}, \qquad 1 = \text{Fully Correct}$$

Manual Evaluation of Correct Answers		**Large Language Model**		
Category	**# Questions**	**ChatGPT$_{5.2}$**	**Claude$_{\text{Opus 4.6}}$**	**Gemini$_{\text{3 Pro}}$**
Data	6	5	5	5.5
Output	10	9	8	6
Performance	5	4.5	4	4
How (global model-wide explanations)	8	6.5	6.5	7
Others	5	5	5	4.5
UI	2	2	1.5	2
Total	**36**	**32**	**30**	**29**

Overall, the models performed well in the manual evaluation section. ChatGPT achieved the highest score with 88.9% accuracy. Claude and Gemini closely followed with 83.3% and 80.6% respectively.
The categories *Others*, *UI*, and *Data* were the most successfully answered by the LLMs, averaging scores of 96.7%, 91.7%, and 86.1% respectively. *Output* was the category the LLMs struggled answering questions for the most, with an average score across the LLMs of 76.7%.

Level of Correctness	**ChatGPT$_{5.2}$**	**Claude$_{\text{Opus 4.6}}$**	**Gemini$_{\text{3 Pro}}$**
Fully Correct	28	26	23
Partially Correct	8	8	12
Incorrect	0	2	1

ChatGPT answered questions the most accurately, with 28 fully correct answers, and the only LLM with no recorded incorrect answers. This is followed by Claude which performed similarly to ChatGPT, with 26 fully correct answers and the same number of partially correct answers (8). Gemini performed the worst with only 23 fully correct answers, and had the highest number of partially correct answers (12).
In total, only 3 incorrect answers were observed, coming from the Claude (2) and Gemini (1). Claude failed to answer the questions "What is the sample size of the training data?", and "What is the system's overall logic?" Displaying different types of incorrectness. In the first instance, it failed to identify an 80:20 train test split, and only provided the dimensions of the overall dataset. For the second question, it referred to variable importance, interpreting the "logic" as the models internal architecture, when instead it needed to see the AI system overall as a prediction tool and explanation interface. Gemini failed to

answer "What are the limitations of the system?" referring to only the poorer recall performance for the positive class (i.e. if the respondent received the H1N1 vaccination). However, this answer relates more closely to the question "What kind of mistakes is the system likely to make?" An ideal answer was to provide information about data limitations and bias, and the fact that the AI system is a static tool without any dynamic learning or updates.

Completeness, Faithfulness, ACF

In this section we investigate the metrics for completeness (i.e. ROUGE-L Recall, BERTScore Recall), Faithfulness (i.e. SBERT), and the constructed ACF score (i.e. equation (∗)).
First, looking at the overall distributions across these metrics. Then, taking a more granular view.

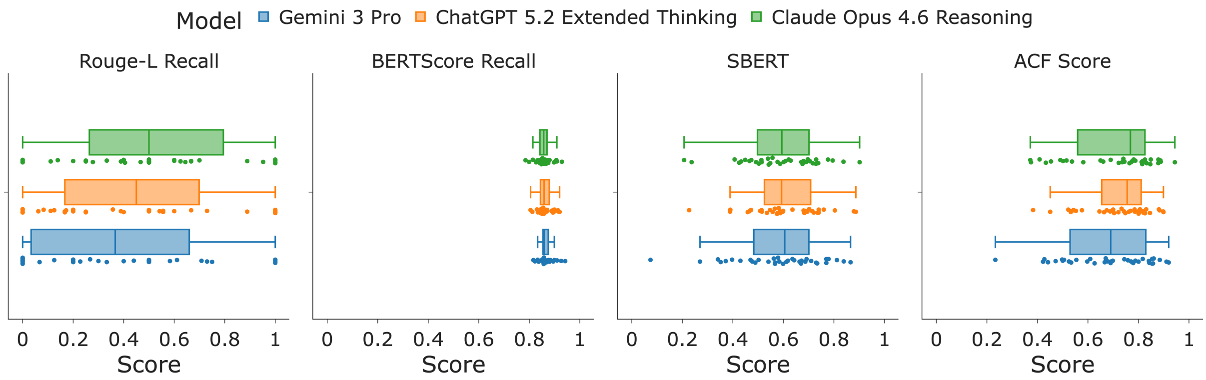


Figure 1: Overall: Boxplot of ACF score and its components metrics for LLM generated answers

Claude has the highest average ROUGE-L recall score of 0.530, ChatGPT comes second with 0.458, and Gemini ranks last with a score of 0.407. The results for BERTScore Recall are inverted in comparison with ROUGE-L Recall, despite both measuring completeness. Gemini produces the highest score of 0.868, followed by ChatGPT with 0.862, and Claude scoring the lowest with 0.857.
Combining these two metrics produces a balanced view. Required values and terms are captured, but overall topics are also preserved.
SBERT is to ensure the LLM generated answers are faithful to their corresponding human answers which are used as reference (i.e. ground-truth/gold standard). This is to measure how well the LLM responses capture the overall meaning of the answers, in terms of their shared semantic similarity. ChatGPT performs the best, with a score of 0.611, followed by Claude scoring 0.593, and Gemini with 0.586.
Entering the manual evaluation phase, Claude is the best overall performer with a partial ACF score of 0.642. ChatGPT is second with 0.635, and Gemini is last with 0.611. All LLMs achieve a respectable level of performance in terms of their completeness and faithfulness, acknowledging that the human answers are subjectively produced.
Full ACF scores were computed for each question using equation (∗) above. The best performing model overall was ChatGPT with an average ACF score of 0.720 across all of the selected XAI-QB questions. Claude was second with 0.706. Gemini ranks third with 0.676.
Splitting this view across categories can also provide valuable information about which types of questions

the LLMs respond better to. In addition to boxplots showing the distribution of data across all points, radar plots are also provided showing the average category performance of LLMs across the component metrics, allowing for clearer comparison.

Figure 2: Split by category: Boxplot of ACF score and its components metrics for LLM generated answers. Average scores summarised below by their corresponding radar plots.

There are some clear insights that can be taken from this analysis. Firstly, ROUGE-L$_{\text{Recall}}$ is on average the lowest for the *Output* category, which infers that not all of the required topics are being covered. This corresponds to the manual evaluation phase where *Output* was the worst performing across the models. Specifically, Gemini with an accuracy score of only 0.6. In terms of their BERTScore$_{\text{Recall}}$, all models perform very well, and equally similar to one another, with minimal intra- and inter- category variance. Claude has a noticeable drop in SBERT *UI* performance compared to ChatGPT and Gemini. This is reaffirmed with its accuracy being the lowest across the LLM models. Indicating it is not faithful to the human reference and what is expected. In terms of the overall ACF score, all LLMs perform comparably well with one another. However, Claude slightly below in *UI*, and Gemini slightly below in *Output*. This is a reflection of their aforementioned poor performance for these categories in the component metrics.

5.2 Readability

Four main measures were selected to assess the readability of XAI-QB answers. The Flesch Kincaid Grade Level, SMOG Index, Coleman Liau Index, and Dale Chall Readability Score. Using a radar plot, we can compare the average readability scores across these metrics for LLM and human generated answers. There is a unified consensus of the rankings of LLMs in terms of their readability. Gemini is the easiest to read with the lowest average score across each metric, followed by ChatGPT, and finally Claude which is the most difficult. In addition to this, human answers beat all of the LLMs in terms of readability with the lowest scores.

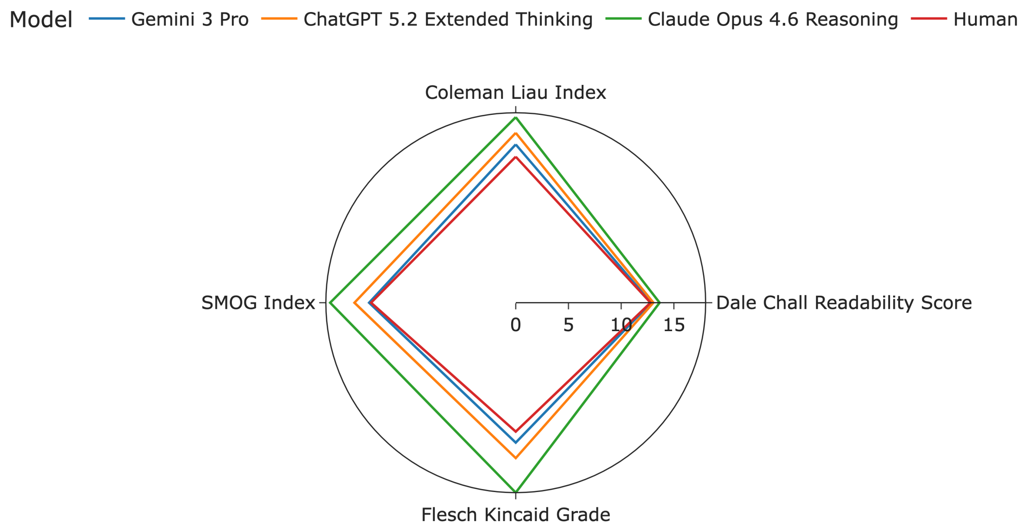


Figure 3: Readability scores across different measures for LLMs and reference human answers.

Model	Human	ΔChatGPT$_{5.2}$	ΔClaude$_{\text{Opus 4.6}}$	ΔGemini$_{\text{3 Pro}}$
Flesch Kincaid Grade	12.22	+2.52	+5.79	+1.05
SMOG Index	13.73	+1.57	+3.88	+0.18
Coleman Liau Index	13.84	+2.26	+3.75	+1.17
Dale Chall Readability Score	12.79	+0.26	+0.86	+0.05

Gemini is most readable of the LLMs with an average level of 13.76, ChatGPT is second with 14.80, and Claude is the most difficult to read with an average level of 16.71.

The purpose of these metrics is not the minimise the readability score. Rather, the human answers provide a baseline for the expected level of reading ease required to answer the questions sufficiently, and understandably. On average, LLMs are more difficult to read, but there will be some variability across these answers where LLMs are more or less readable than their corresponding reference human answers. Therefore, we look at the distribution of the difference between human and LLM readability metrics. In other words

$$\text{Difference } (\Delta) = \text{Readability}_{\text{LLM}} - \text{Readability}_{\text{Human}}$$

This gives a distribution centred around the human answer's readability for all 36 questions. A distribution centred on 0 would mean on average the readability is equivalent to humans. A distribution shifted to the left/right of 0 is easier/harder to read on average than their corresponding human answers.

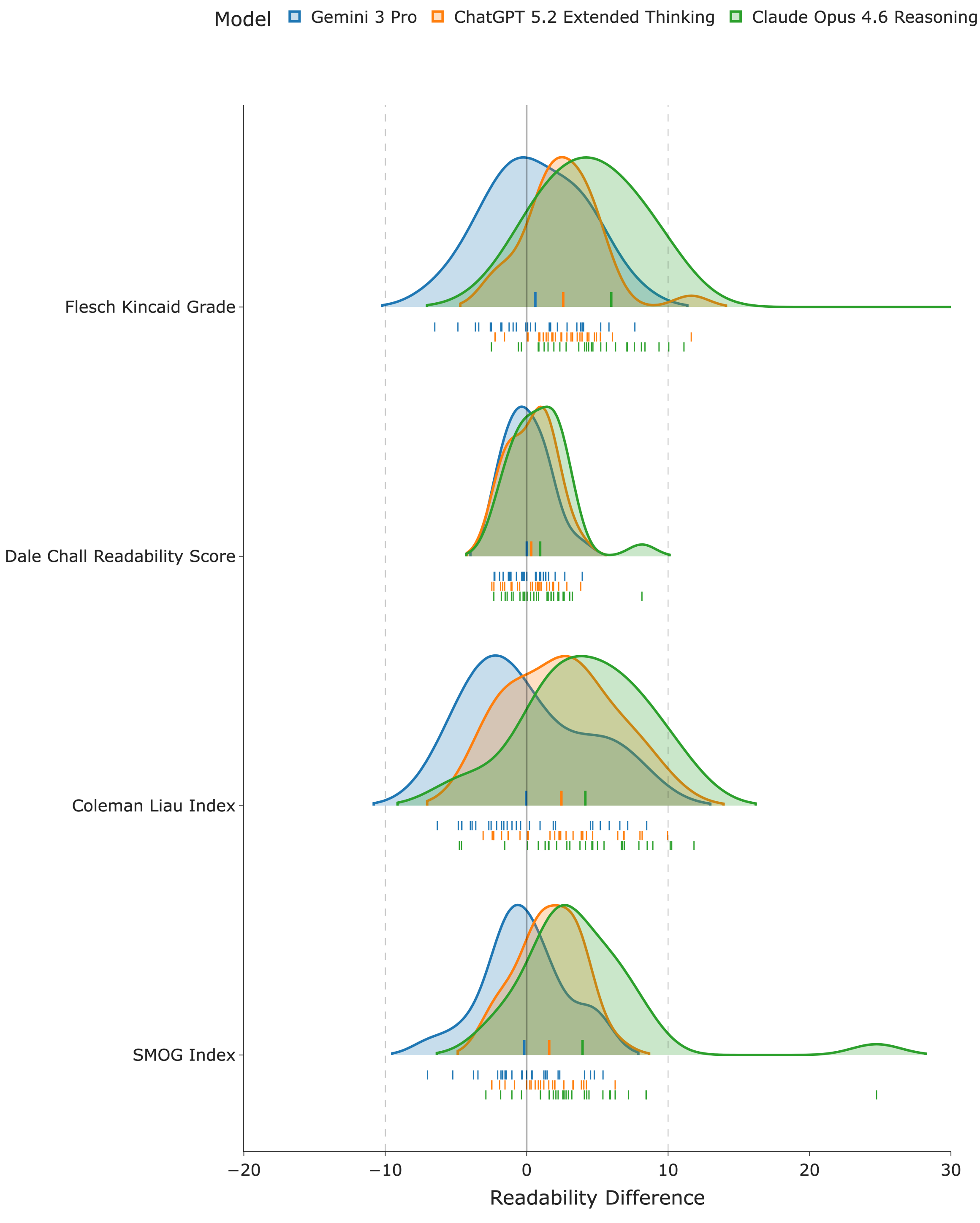


Figure 4: Distribution of LLM readability relative to reference human answers (i.e. difficulty delta)

The distributions for each LLM across these various metrics display the same general shape in terms of their skewness and kurtosis. All of the LLMs mean distribution values are shifted to the right of 0, which reaffirms on average LLMs answers are more difficult to read compared to the corresponding human answer. However, a significant number of answers from Gemini are easier to read than human answers, with a noticeable right-skewed distribution compared to ChatGPT and Claude. Conversely, Claude has a noticeable left-skewed distribution compared to the other LLMs, meaning a significant proportion of its answers are less readable than its comparative human answers. ChatGPT has the smallest variance, displaying the ability to answer questions at a more consistent readability level compared to Gemini or Claude. Claude also has a selection of outliers in the distribution with significantly higher (i.e. more difficult) readability scores. These will be discussed in the following section.

Investigating the responses further, average word and sentence counts are computed, and difficult words are identified. A difficult word is determined to be any word which is three syllables or greater. These summary statistics allow the measurement of how verbose certain LLMs are compared with others.

Metric / Model	**ChatGPT**$_{5.2}$	**Claude**$_{\text{Opus 4.6}}$	**Gemini**$_{\text{3 Pro}}$	**Human**
Sentence count	2.92	3.58	**1.97**	3.03
Word count	63.67	91.17	**37.17**	60.78
Words / sentence	22.01	29.32	**19.55**	21.41
Difficult words	15.00	24.39	**7.22**	13.17

Interestingly, Gemini performs the best on all of these statistics, even compared with the human answers. This includes the fewest number of difficult words on average. However, this does not translate into an easier level of readability. Claude is noticeably the most verbose, responding to answers with more sentences of greater length compared to ChatGPT and Gemini.

5.3 In-Depth Investigation: ACF and Readability Extremes

Looking at the LLM generated responses which achieved the highest and lowest scores for ACF and readability, this will give us an insight into the the types of questions that are more or less interpretable by the LLMs.

ACF

Certain categories contain questions which are more open to interpretation than others, and therefore will deviate more from the reference human answer. This is shown by looking at the average ACF scores by LLM across categories.

Category	**Average ACF score**			
	ChatGPT$_{5.2}$	**Claude**$_{\text{Opus 4.6}}$	**Gemini**$_{\text{3 Pro}}$	**Average**
Data	0.718	0.778	**0.781**	0.759
Output	**0.686**	0.644	0.563	0.631
Performance	**0.739**	0.691	0.674	0.702
How (global model-wide explanations)	0.701	**0.702**	0.667	0.690
Others	0.773	**0.796**	0.734	0.767
UI	0.787	0.631	**0.817**	0.745

The categories *Others* and *Data* have the highest ACF scores on average across all of the LLMs, with 0.767 and 0.759 respectively. These questions are noticeably more straightforward and clearly defined, relating to the AI system's context, as well as the source and specifications of the data. These can be easily located and inferred from the source code provided.
The category *Output* on the other hand has the lowest ACF score, with an average of 0.631 across all of the LLMs. These questions can be open-ended, such as how the AI system's output is best utilised, or how it fits into a workflow. Additionally, the "output" may be interpreted differently. The LLM may determine this to be only one aspect of what's produced in totality, such as the predictions, global explanations, or What-If? dashboard. These are all valid outputs, and should therefore all be considered.
Average ACF score for LLMs give us a ranking of the questions which are best interpreted by all of the LLMs.

Category	**Question**	**Rank**	**Average ACF score**
Data	What is the source of the training data?	1	0.909
How (global)	What kind of algorithm is used?	2	0.889
	...	...	...
How (global)	What is the system's overall logic?	35	0.466
How (global)	How does it weigh different features?	36	0.441

We will focus on the questions with the highest/lowest ACF score for further insight, selecting the answer from the LLM which provided the best/worst ACF score for this question.

Category	Data
Question	What is the source of the training data?
Model	**Claude**$_{\text{Opus 4.6}}$

ROUGE-L Recall	**Bert Recall**	**SBERT**	**Accuracy**	**ACF score**
0.952	0.909	0.902	1	0.944

Information units: ["DrivenData", "Flu Shot Learning: Predict H1N1 and Seasonal Flu Vaccines", "US National 2009 H1N1 Flu Survey (NHFS)"]
Human Answer: "Data is sourced from DrivenData's 'Flu Shot Learning: Predict H1N1 and Seasonal Flu Vaccines' competition and originates from the US National 2009 H1N1 Flu Survey (NHFS). This survey was conducted between October 2009 and June 2010 as a one-time telephone survey of households to monitor vaccinations during the 2009-2010 flu season in response to the 2009 H1N1 pandemic."
LLM Answer: "The training data originates from the National 2009 H1N1 Flu Survey (NHFS), which was sponsored by the National Center for Immunization and Respiratory Diseases (NCIRD) and conducted jointly by NCIRD and the National Center for Health Statistics (NCHS) at the Centers for Disease Control and Prevention (CDC). The NHFS was a list-assisted random-digit-dialling telephone survey of households in the United States, conducted between October 2009 and June 2010. The dataset was accessed via DrivenData's public competition 'Flu Shot Learning: Predict H1N1 and Seasonal Flu Vaccines'"

This answer aligns very well with both the information units and the reference human answer, as shown across all metrics. It's accurate, covers all of the points required, and it is faithful to how the human answer was written.

Category	How (global model-wide explanation)
Question	How does it weigh different features?
Model	**Claude**$_{\text{Opus 4.6}}$

ROUGE-L Recall	Bert Recall	SBERT	Accuracy	ACF score
0	0.834	0.207	0.5	0.375

Information Units: ["Split nodes", "Reduce error (loss function)", "Gain"]
Human Answer: "XGBoost splits nodes by maximising gain, which measures how much a split reduces the prediction error (loss function). Each feature's usefulness is measured by how often and how effectively it is chosen to split nodes during tree construction. Features that produce greater reductions in loss get selected more often. Internally, each split's 'gain' (improvement in the objective) is stored - this can be viewed as a weight for that feature."
LLM Answer: "Variable importance analysis reveals that the system assigns the highest importance to: opinion on H1N1 vaccine effectiveness, doctor recommendation for H1N1 vaccine, and opinion on H1N1 risk. These opinion and recommendation features have substantially higher importance scores than other variables. Features grouped as 'opinion' and 'doctor recommendation' categories collectively have the greatest impact on model performance. Behavioural features (e.g. hand washing, mask use) and protected demographic features (e.g. age, sex, race) have comparatively lower individual importance, though they still contribute to the overall prediction. Geographic and household variables have the least influence."

This answer interprets the question completely differently to the reference human answer, shown most significantly by the low ROUGE-L and SBERT scores. The reference considers the internal architecture of the XGBoost classifier model in terms of its nodes and how they're split. The answer by the LLM however refers to the variable importance plots. This misinterpretation most likely comes from the ambiguity about what "it" refers to in the question. Furthermore, the LLM fails to identify there is another XAI-QB question "What are the top rules/features that determine its predictions?" which is more suited to talking about variable importance.

Readability

Similar analysis is performed for readability.

Category	Average Readability Score			
	ChatGPT$_{5.2}$	Claude$_{\text{Opus 4.6}}$	Gemini$_{\text{3 Pro}}$	Average
Data	**13.67**	16.17	14.09	14.64
Output	15.72	16.01	**13.54**	15.09
Performance	13.46	15.39	**11.23**	13.36
How (global model-wide explanations)	**15.35**	16.90	16.38	16.21
Others	14.24	15.32	**13.27**	14.28
UI	16.09	27.91	**10.87**	18.29

The category *Performance* has the easiest readability, whereas the category *UI* is significantly harder to read compared to other categories. However this is largely inflated due to Claude's answers for these questions.

Category	Question	Rank	Average Readability Score
Others	Does the system learn specifically from my interaction?	1	11.31
Performance	How accurate/precise/reliable are the predictions?	2	11.56
...	...	...	...
How (global)	What features does the system consider?	35	19.14
UI	What does this UI element do?	36	19.79

Category	Others
Question	Does the system learn specifically from my interaction?
Model	**Gemini**$_{\text{3 Pro}}$

SMOG Index	Coleman Liau Index	Dale Chall Readability Score	Flesch Kincaid Grade
8.84	8.21	10.45	7.78

LLM Answer: "No, the system does not learn from individual user interactions. It uses a pre-trained model saved in a 'pickle' file."

This is a very short answer which does not include many difficult words. The use of the word "pickle" however is deceptive, as this requires data science knowledge outside the scope of a healthcare practitioner. This violates the guidelines set in the instructional template.

Category	UI
Question	What does this UI element do?
Model	**Claude**$_{\text{Opus 4.6}}$

SMOG Index	Coleman Liau Index	Dale Chall Readability Score	Flesch Kincaid Grade
35.95	19.72	19.95	58.32

LLM Answer: "The system's interface includes several key visual elements: variable importance bar charts display features ranked by their impact on model performance (measured by dropout loss — higher values indicate greater importance); partial dependence profile plots show how the average predicted probability changes as a single feature varies across its range, helping users understand the direction and magnitude of each feature's effect; accumulated local effects plots provide a similar view but account for correlations between features, offering a more robust interpretation; break-down plots decompose an individual prediction into positive and negative contributions from each feature, showing which factors pushed the prediction higher or lower; and the what-if dashboard presents an interactive scatter plot where users can select individual data points, modify their feature values, and observe real-time changes in predicted probabilities."

The average readability of this answer is inflated largely due to its SMOG and Flesch Kincaid Grade values. SMOG looks at polysyllabic count (i.e. words having more than one syllable), of which this answer has 33 (compared to an average across all LLM generated answers of only 15.6). Flesch Kincaid (or at least `textstat`) does not consider ";" to be the end of a sentence. Therefore, the total number of sentences is evaluated to be much smaller, which increases the (total words/total sentences) term in the equation, and thus the total score. These two factors inflate SMOG and Flesch-Kincaid metrics, resulting in the answer being evaluated as more difficult to read.

Average scores for ACF and readability, along with their rankings, can be found in the Appendix.

6 Discussion

As shown above, LLMs can answer XAI-QB questions sufficiently well when measured against reference human generated answers. Performing to a strong degree across the components of accuracy, completeness, faithfulness, and readability. This is particularly difficult due to the technical and sometimes open-ended nature of the questions. A selections of these answers can be inferred easily from the source code provided, whilst others require a more nuanced contextual understanding, taking a wider view of how the AI system functions. Furthermore, the subjectiveness of the human explanations and the strict requirements to cover all relevant topics posed a challenge for the LLMs. In all of these circumstances, the LLMs proved to be capable at handling these challenges. This validates that LLMs can indeed be used as a conversational explanatory agents, supplementing explanations and providing personalised guidance within an explanation interface.

Accuracy

Requiring manual evaluation, the accuracy of an LLM generated response is the most important, yet time consuming metric to evaluate. This metric detects LLM hallucinations, ensuring the answers are relevant and factually correct. This is crucial when deploying an AI system into a high-stakes domain where the answers would be used to aid critical decision-making processes.

In total, 71.3% of answers were graded as fully correct across the LLMs, with a further 25.9% being graded as partially correct. This means that overall 97.2% of answers provided by the LLMs displayed a varying level of correctness when manually evaluated. Only 3 questions were incorrect, none of which were the result of hallucinations. These incorrect answers were due to either a misinterpretation of the question, or a failure to identify the relevant section of source code. There was no observed instance where false or misleading information was presented as fact.

This evaluation suggests that when applied to high-stakes domains, LLMs are capable of accurately explaining an AI system across the XAI-QB categories: *Data*, *Output*, *Performance*, *How (global model-wide explanations)*, *Others*, and *UI*. Noticeably, categories which contain more specific questions results in a higher accuracy (such as *Others* (96.7%), *UI* (91.7%), and *Data* (86.1%)). These questions have explicit answers associated with them, and can be located straightforwardly within the source code. Categories with vague or open-ended questions have lower accuracy (such as *Output* (76.7%)), as this introduces a level of subjectivity and interpretation to the answers.

Partial correctness can provide useful information about the AI system itself. Highlighting the vulnerabilities or limitations in the AI system that need to be addressed. In particular, a proportion of these answers failed to recognise the AI system as the complete prediction tool and explanation interface, rather choosing to focus on one particular element. This is shown in the lower *Output* accuracy, where responses either spoke solely about the predictions, or the explanation interface. This suggests a lack of cohesion across the AI system elements, and an unclear design practice. This is indeed true, as the source code does not clearly state what is accessible to the user.

A suggested improvement would therefore be to utilise the XAI-QB more as a framework, for which open-ended questions are clarified and refined. By adding a required level of detail, based on the AI system's specific use-case and application, a more complete, relevant, and accurate answer would be expected. This can be fine tuned in the instructional template and further evaluated.

Completeness

Completeness is a necessary metric to ensure all required topics are being discussed in an answer. This is crucial in high-stakes domains, where decisions need to be fully informed and users have access to all of the relevant information. Incomplete answers allow unresolved gaps in knowledge and understanding, which may result in severe consequences when making critical decisions.

ROUGE-L scores are expectedly strict, as it looks for the longest common subsequence of our information units in the LLM generated response. Information units were constructed to be flexible, but as ROUGE-L uses exact word matching in the same subsequence order, this means it is very sensitive to how answers are worded. As a result, ROUGE-L has the widest distribution and lowest average value of our ACF score components, spanning across the entire [0, 1] interval, and recording an average score of 0.465 across the LLMs. This means ROUGE-L can find the exact terminology outlined by the information units in the LLMs responses just under half of the time.

Output has the lowest average ROUGE-L score. This corresponds to the manual evaluation phase where *Output* was the worst performing category across the models, reaffirming the open-endedness of the questions causing the answers to deviate more significantly from reference human answers. Expectations for these questions need to be clarified. The *How (global model-wide explanations)* category has the widest distribution, which also implies a selection of questions are ill-defined and require further clarification. Indeed this category contains a mixture of clearly stated (e.g. "How does the system make predictions", "What kind of algorithm is used") and vague questions ("What is the system's overall logic", "What kind of rules does it follow?").

All LLM responses with a high ROUGE-L score can be considered complete. However, not all LLM responses with a low ROUGE-L score can be considered incomplete.

BERTScore was introduced to validate answers which capture the information units, but stated differently, such as synonyms, paraphrases and distant dependencies. In contrast to ROUGE-L, BERTScore has the narrowest distribution and highest average value of 0.862 for the ACF score components. This implies highly similar candidates for the information units are frequently found in the LLMs answers, inferring a higher degree of completeness compared to its corresponding ROUGE-L score. Furthermore, the small inter- and intra- model variance suggests the LLMs are all answering the questions using different wording and terminology, but covering the same required topics.

Combining these two metrics $\frac{1}{2}\left(\text{ROUGE-L}_{\text{Recall}} + \text{BERTScore}_{\text{Recall}}\right)$ gives a more balanced view of completeness. Ensuring that key terminology and specific values are being referenced, whilst also allowing a degree of flexibility with how these answers are phrased.

Overall, the LLMs averaged a completeness score 0.663, suggesting the answers are covering most of the required topics, with some absences. This can somewhat be attributed to the subjectivity of the information units and the strict nature of the ROUGE-L score. Accuracy and completeness are also closely related. Looking at the average scores across categories, rank(accuracy) $\approx$ rank(completeness). This aligns with expectations, where fully correct answers are more complete than partially correct answers. This completeness score is also influenced by the LLMs level of verbosity, which will be discussed in a subsequent section below.

Faithfulness

Completeness evaluates whether the correct terminology is included in an LLMs response, but it does not assess *how* it's being used. A response may include the right terminology and appear plausible, but does not faithfully reflect the internal reasoning and logic of the AI system it is describing. This is crucial in high-stakes domains, where an unfaithful answer could influence critical decisions in the wrong direction, and provide misleading justifications.

SBERT looks at the semantic similarity of an LLMs answer when compared to its corresponding reference human answer. This is to evaluate if the desired meaning of the answer is being properly conveyed. With an average score of 0.596 across the LLMs, this indicates a moderate semantic alignment between LLM and human reference answers. While the LLMs capture the general intent of responses, they diverge in terms of detail and structural approach to an answer. This expected due to the inherent subjectiveness of the reference human answer. The categories *Data* and *UI* have the highest average SBERT scores across the LLMs. These categories have clearer and more specific answers associated with their questions, due to a limited number of ways these answers can be formulated. As a result, they do not deviate as significantly from the reference human answers. Notably however, Claude underperforms in the *UI* category compared to ChatGPT and Gemini, with a significantly lower SBERT score. This reconciles with Claude's accuracy score for *UI*, being the only model not to register fully correct answers for all of the questions.

ACF score

The constructed ACF score, formulated by equation $(*)$, combines all of these component metrics into a single score. This score aims to evaluate the overall quality of an answer provided by the LLMs in relation to reference human answers and its information units. All of these components are equally weighted to give a holistic measure for how an LLM performs as an explanatory conversational agent.

The ACF score is validated by the empirical evaluation of best/worst performing answers. Generally, the rank of the ACF score reflects the open-endedness of the question. As the level of interpretation allowed by a question increases, there is a greater deviation from the ground-truth answer (provided by the human reference and information units). The highest scoring questions all have specific answers that can be found directly in the source code, and do not require detailed or lengthy explanations. This means the answers can be answers easily identified, and answered succinctly. This reduces the variability and therefore increases the ACF score. The worst performing questions are more abstract, and require the LLMs to infer contextual nuance. This means there will be a greater level of variability and a deviation away from reference human answers.

This shows that the equation serves as a useful benchmark for quantitatively analysing the similarity of an LLM's answer with its corresponding reference human answer.

Readability

In addition to being accurate, complete, and faithful, LLMs acting as conversational explanatory agents must produce answers which are readable. This ensures that the information being presented is accessible and understandable to the user relative to their domain and use-case. Setting a baseline for readability, the reference human answers are the easiest to read, with an average grade level across the readability metrics of 13.14. The average readability score across LLMs is slightly higher with a grade level of 15.09 ($\Delta + 1.95$), suggesting that LLMs are not able to reduce down complex concepts and terminology into simpler terms as effectively as the human generated answers. Comparing LLMs, Gemini produces answers which are the easiest to read, with an average grade level of 13.76 ($\Delta + 0.61$). Claude produces answers which are the most difficult to read, with an average grade level of 16.71 ($\Delta + 3.57$). This reflects the difference in writing styles, with Gemini producing fewer, shorter sentences, while Claude generates lengthier answers with more complicated language (3 times as many difficult words on average compared to Gemini). However, there is a noticeable trade-off between readability and accuracy, highlighted by Gemini producing the highest proportion of partially correct answers. For instance, looking at the example:

Category	Output
Question	What is the amount of the output data?
Model	**Gemini**$_{3\ \text{Pro}}$

Average Readability: 14.61 ($\Delta - 0.16$) **Accuracy**: 0.5

LLM Answer: "For every patient profile input, the system returns a single probability score and a single binary classification label."

Model	**Claude**$_{\text{Opus 4.6}}$

Average Readability: 19.12 ($\Delta + 4.35$) **Accuracy**: 1

LLM Answer: "For each individual respondent, the system produces a single predicted probability value and, optionally, a binary classification. The explainability components produce additional outputs: a ranked list of variable importance scores across all features, partial dependence and accumulated local effects profiles for each feature, and individual-level break-down explanations showing how each feature contributed to a specific prediction. The what-if dashboard produces updated predictions for modified scenarios on a per-respondent basis."

Despite Gemini's answer being scored as much more readable, it fails to answer in the required detail, providing an incomplete assessment that the output data only relates to the predictions generated. Claude correctly identifies that in addition to this, the AI system produces explainability plots and a What-If? dashboard.

Model benchmarking

All models appear to excel at different components of the ACF score, providing interesting insights into how these LLMs construct their answers, and how these align with the reference human answers and information units.

ChatGPT has the highest accuracy and faithfulness scores, providing the correct information in a semantically meaningful way. Claude has the highest ROUGE-L and overall completeness score, showcasing its ability to cover all of the required material. However, Claude produces the most verbose answers which means there is a higher chance these information units will be captured by ROUGE-L and BERTScore. Therefore, this score may be inflated by virtue of providing longer sentences compared to ChatGPT and Gemini. Gemini has the highest BERTScore and easiest readability scores, showcasing an ability to answer comprehensively yet succinctly. However, there is a trade-off with accuracy. Gemini is the least accurate of the LLMs, which may be due to shorter answers producing much more partially correct answers.

Gemini has highest ACF scores for *Data* and *UI*. These are categories more suited to shorter answers, as the questions have well defined answers which can be located easily in the AI system's source code. ChatGPT has the highest ACF scores for *Output* and *Performance*. Output was the most challenging category for the LLMs to answer correctly. As ChatGPT was the most accurate, this is expectedly reflected in the overall ACF score. A similar assessment can be made for *Performance*. The questions within these categories are more open-ended, which infers ChatGPT is the best model suited to handling contextual nuance, and answer in line with human reference answers when questions require a greater level of interpretation. Claude has the highest ACF scores for *How (global model-wide explanations)* and *Others*. *How (global model-wide explanations)* requires a lot of descriptive text in order to successfully cover all aspects of the answer. This favours the longer answers of Claude, as there is more detail that is required to successfully answer these questions.

ChatGPT is the highest performing model in terms of overall ACF score. Producing the most consistent results across all dimensions of accuracy, completeness, and faithfulness. Furthermore, ChatGPT provides these answers in a consistently accessible way, with the smallest variance in readability scores. With the ability to provide accurate answers, in a semantically meaningful way, which are accessible to the user, ChatGPT offers the best answers overall in terms of explainability within a high-stakes domain. Gemini proves useful for quick, easily digestible answers, which is less suited for high-stakes domains where accuracy of information is essential. Claude is more considered with its longer, yet more technical answers, which may prove too challenging for those without AI/ML domain knowledge. ChatGPT achieves a suitable balance between all of these aspects.

Challenges and Additional Comments

These LLMs have showcased their ability to generate high quality answers, quickly. In contrast, producing reference human answers is time consuming and took comparatively much longer. Manual evaluation of answers was additionally time consuming, and isn't feasible for all the runs produced by ChatGPT, Claude, and Gemini (totalling 324 answers to evaluate). However, generating reference human answers and manual evaluation is necessary in order to provide a level of oversight and accountability that is in line with HC-AI principles and design practices. Filtering runs for manual evaluation based on partial ACF score was validated when it is observed that rank(partial ACF score) = rank(accuracy). This signifies that using the partial ACF score preserves the order and can infer the relative level of correctness when comparing LLMs. Considerations also need to be made for clinical safety and misuse when LLMs are being used as conversational explanatory agents. This occurs when a disproportionate amount of trust is granted to explanations that are readable but not accurate. The fluency and readability of an answer may create a false impression of plausibility for incorrect answers, which cannot be validated due to a clinical practitioner's lack of technical knowledge. This vulnerability reinforces the need for the ACF framework to be used in conjunction with readability analysis, to ensure factually correct and complete information is being presented in an understandable manner. Clinical adoption should be contingent on all of these metrics working together, supplemented by human-in-the-loop verification.

Limitations and Future Work

There are certain limitations of this project when validating LLMs as conversational explanatory agents. As API access was restrictive and model parameters could not be explicitly set (i.e. `temperature`, `top_p`, and `max_tokens`), reproducibility must be empirically assessed rather than guaranteed by design. The evaluations across each of the LLM's three runs demonstrated minimal variance across component metrics, providing evidence of sufficient output stability under these constraints. However, this protocol should be reapplied with each future model release or platform update, as changes to underlying model weights or inference behaviour may affect response consistency (if API access restrictions persist). This research considers a narrow use-case within the domain of healthcare, investigating only one AI system applied to a single source of data. Analysis was conducted on a subset of the XAI-QB, excluding any questions about local, prediction level behaviour (i.e. *Why*, *Why not*, *How to be that*, *How to still be this*, *What If* categories). Furthermore, there are no questions in the XAI-QB which relate to diversity, inclusion, and fairness, which are key components of HC-AI for reliable, safe, and trustworthy systems. [78, 57, 42, 39]. When measuring completeness and faithfulness, these comparisons were made against one set of reference human answers. This evaluation can be expanded by including additional questions (e.g. local prediction level behaviour, ethical considerations), and made more robust (i.e. less subjective) by using multiple sets of reference human answers as ground-truth labels. Future work may consider testing AI systems across various different domains, datasets, and use-cases to get a broader and more complete view of how LLMs can be used as conversational explanatory agents. Clinicians and public health professionals can be included in human-grounded evaluation and user studies, to evaluate qualitative measures such as user satisfaction, comprehension, trust, and acceptance. The completeness component (and therefore the overall ACF score) is sensitive to verbose answers, where there is a greater chance for information units to be captured. Therefore an obvious immediate improvement would be to introduce a penalty for lengthier answers. The ACF score can also be evaluated for different objectives by altering the weights of certain component metrics. In a high-stakes domain, accuracy and strict completeness (i.e. ROUGE-L), could be weighed higher to ensure all factually correct information is being presented. Alternatively, different composite scores or uncertainty metrics can be constructed to evaluate LLM generated responses when compared to reference human answers. The instructional template can also be further refined or altered, based on desired objectives, domains, target audiences, and questions. This should be done with caution, as the fine-tuning is empirically not as influential as expected (e.g. developing a persona, adjusting the guidelines). Using reference human answers as a mark scheme, LLM-as-a-judge [91] —an evaluation method where LLMs are used to automatically assess the outputs from other AI models —could be investigated as a method to expedite the process of evaluating accuracy (doing so for all runs and not a selection).

7 Conclusion

This research project aimed to evaluate the answers of different LLMs (**ChatGPT**$_{5.2}$, **Claude**$_{\text{Opus 4.6}}$, **Gemini**$_{\text{3 Pro}}$) when given the task of explaining an AI system. This AI system simulates a real world scenario where AI-assisted decision-making is applied to the setting of a high-risk domain (i.e. healthcare), acting as a prediction tool and explanation interface. In order to enhance trust and increase the adoption of such systems, these LLMs were deployed as conversational explanatory agents to elucidate user understanding and fill in knowledge gaps.

The questions were sourced from the extended XAI-QB, spanning the entire AI framework and covering aspects of data, output, performance, global model-wide explanations, system functionality, and UI. The AI system was designed and developed to specifically answer these questions, and was composed of files for exploratory data analysis, modelling, explanations, and a What-If? dashboard. Each file was properly documented and produced insights which fed into the next. In conjunction, reference human answers and information units were produced for these questions. These served as the ground-truth labels to measure against the LLMs responses, assessing their levels of *accuracy*, *completeness*, and *faithfulness*. The metrics were analysed individually, and combined into an overall ACF score which allowed for the holistic evaluation of LLM generated answers, and for future comparisons across different LLMs and AI system applications. Furthermore, the reference human answers provided a benchmark for *readability*, setting a baseline for the required level to sufficiently answer technical questions in a comprehensible way, balancing theoretical detail with understandability.

The LLMs were prompted with an instructional template, consisting of metadata, persona, instructions, and XAI-QB questions. This was provided in a JSON format to ensure that the instructions were being strictly followed, and the output was provided in a consistently structured format. The instructional template was designed to be iteratively fine-tuned, based on successive prompting and evaluation.

Overall the models performed strongly, with an average accuracy of 0.843, completeness of 0.663, and faithfulness of 0.596. This combines into an overall average ACF score of 0.700, showcasing the ability to answer XAI-QB questions correctly, covering most of the topics required, in a style which follows a general semantic similarity to the reference human answers. This is a particularly strong performance considering the subjective nature of the reference human answers (and to a lesser extent, the information units). This displays the LLMs clear utility as conversational explanatory agents when tasked with explaining an AI system. These LLMs exhibited different strengths and weaknesses, highlighting their specific proficiencies and suggesting optimal use-cases for where they can be applied.

ChatGPT performed the best overall, displaying an ability to answer the most accurately, and interpret questions more in line with the reference human answers. Furthermore, the writing style of ChatGPT was the most consistent, with smaller variance in its readability level across all answers. Claude is more proficient at producing in-depth analytical answers, which is suited to a more technical audience where detail is preferred. Gemini produces the most accessible answers, with the easiest readability level and shortest length of answers. This is suited for synopses and summaries of the AI system.

Finally, the ACF score was able to successfully identify questions from the XAI-QB which required further refinement and clarification. Lower ACF scores were aligned with more open-ended, vague questions which would benefit from additional details. Furthermore, the ACF score also assists in highlighting areas of the AI system which lacked coherence and clear design practices. This is the case when the LLMs provide correct, but irrelevant information.

Altogether, this work demonstrates that the structured evaluation using the ACF framework with reference human answers, information units, and readability benchmarking is an informative approach to bridging the gap between LLM generated explanations, and the rigorous evaluation techniques needed to validate them.

Appendix

XAI-QB Selected Questions

Data

Question	Answer Source
What kind of data was the system trained on?	eda
What is the source of the training data?	eda
How were the labels/ground-truth produced?	eda
What is the sample size of the training data?	eda
What dataset(s) is the system NOT using?	eda
What are the potential limitations/biases of the data?	eda, explanations

Output

Question	Answer Source
What kind of output does the system give?	modelling, explanations, what-if
What does the system output mean?	modelling, explanations, what-if
What is the scope of the system's capability? Can it do...?	modelling, explanations, what-if
How is the output used for other system component(s)?	eda, modelling, explanations, what-if
How should I best utilize the output of the system?	explanations, what-if
How should the output fit in my workflow?	explanations, what-if
What is the source of the information object(s)?	eda, modelling, explanations, what-if
What is the scope of the output data?	modelling, explanations, what-if
What is the amount of the output data?	modelling, explanations, what-if
What logic is used for the initial output / recommendation?	modelling, explanations, what-if

Performance

Question	Answer Source
How accurate/precise/reliable are the predictions?	explanations
How often does the system make mistakes?	explanations
In what situations is the system likely to be correct/incorrect?	explanations
What are the limitations of the system?	explanations
What kind of mistakes is the system likely to make?	explanations

How (Global Model-Wide Explanation)

Question	Answer Source
How does the system make predictions?	modelling
What features does the system consider?	eda, modelling, explanations
What is the system's overall logic?	modelling, explanations, what-if
How does it weigh different features?	modelling, explanations
What kind of rules does it follow?	modelling
What are the top rules/features that determine its predictions?	explanations
What kind of algorithm is used?	modelling
How were the parameters set?	modelling

Others

Question	Answer Source
How have updates affected the system?	eda, modelling, explanations, what-if
Does the system learn specifically from my interaction?	eda, modelling, explanations, what-if
Why is the system using or not using a given algorithm / feature / rule / dataset?	eda, modelling, explanations, what-if
What are the results of other people using the system?	eda, modelling, explanations, what-if
Who is responsible for this system? Who are the authors?	eda, modelling, explanations, what-if

UI

Question	Answer Source
What are the affordances of the system?	what-if
What does this UI element do?	what-if

Instruction Template

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{
  "metadata": {
    "template_name": "Healthcare XAI Question Bank Instructional Template",
    "version": "1.3",
    "last_updated": "2025-01-25",
    "authors": "Henry El-Jawhari, Robert Koch Institute (RKI), Freie Universität
    ↪  Berlin",
    "description": "An instructional JSON template for a Large Language Model to
    ↪  generate structured, clinically relevant explanations about an AI system for a
    ↪  healthcare audience, based on the eXplainable AI (XAI) Question Bank."
  },
  "persona": {
        "persona": "You are an Explainable AI specialist communicating with healthcare
        ↪  professionals about the safe, transparent, and ethical use of an AI system
        ↪  in clinical and public health settings.",
        "target_audience": "Healthcare professionals (e.g. clinicians, medical
        ↪  researchers)",
        "tone": "Maintain a professional, trust-building tone suitable for clinical
        ↪  communication. It should be factual, clear, and safety-conscious."
  },
  "instructions": {
        "input": [
                "You will be provided with four files that together describe the AI
                ↪  system: 'eda_basic', 'modelling_basic', 'explanations_basic', and
                ↪  'what_if_basic.'",
                "Each file covers a different aspect of the system: exploratory data
                ↪  analysis, model design and training, explainability methods, and
                ↪  what-if analysis.",
                "Treat these files collectively as your sole source of factual
                ↪  knowledge about the AI system. Do not infer or fabricate details
                ↪  beyond what they contain.",
                "Do not reference these files by name in your answers. Present the
                ↪  information as unified knowledge about the AI system."
                ],
        "guidelines": [
                "Use clinically contextualised language. Explain model behaviour,
                ↪  data, and performance in terms of patient care, public health, and
                ↪  clinical decision-making.",
```

```
            "Avoid generic data science or technical phrasing. Translate technical
            ↪  AI/ML terminology into healthcare-relevant concepts (e.g. 'false
            ↪  negative rate' into 'how often the AI system misses a true
            ↪  case').",
            "Prioritise patient safety, ethical transparency, and clinical
            ↪  applicability in every explanation.",
            "Use examples and analogies drawn from healthcare practice (e.g.
            ↪  patient triage, diagnostic reasoning, treatment planning).",
            "Where the input files lack sufficient detail to answer a question
            ↪  fully, state what is known and explicitly flag what information is
            ↪  missing. Do not speculate.",
            "Do not include code, citations, or mathematical derivations unless
            ↪  explicitly relevant to clinical interpretation.",
            "Each answer must help a healthcare professional understand how to
            ↪  safely and effectively use or evaluate the AI system in a clinical
            ↪  or public health context."
        ],
        "output":
        [
            "The answers should be written in British English",
            "Fill in all of the empty string placeholders with the specific
            ↪  details of the AI system.",
            "The 'question_interpretation' string should restate the question in
            ↪  plain language, clarifying what it is really asking.",
            "The 'answer' string should provide a complete answer to the question
            ↪  based on the input files.",
            "Output should be a single JSON file containing only the 'questions'
            ↪  portion of this instruction template.",
            "Do not ask any questions or include conversational text in the
            ↪  output."
        ]
    },
    "questions": {
        "Data": {
            "What kind of data was the system trained on?": {"answer": ""},
            "What is the source of the training data?": {"answer": ""},
            "How were the labels/ground-truth produced?": {"answer": ""},
            "What is the sample size of the training data?": {"answer": ""},
            "What dataset(s) is the system NOT using?": {"answer": ""},
```

```
        "What are the potential limitations/biases of the data?": {"answer":
        ↪  ""}
    },
    "Output": {
        "What kind of output does the system give?": {"answer": ""},
        "What does the system output mean?": {"answer": ""},
        "What is the scope of the system's capability? Can it do...?":
        ↪  {"answer": ""},
        "How is the output used for other system component(s)?": {"answer":
        ↪  ""},
        "How should I best utilize the output of the system?": {"answer": ""},
        "How should the output fit in my workflow?": {"answer": ""},
        "What is the source of the information object(s)?": {"answer": ""},
        "What is the scope of the output data?": {"answer": ""},
        "What is the amount of the output data?": {"answer": ""},
        "What logic is used for the initial output/recommendation?":
        ↪  {"answer": ""}
    },
    "Performance": {
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        "How often does the system make mistakes?": {"answer": ""},
        "In what situations is the system likely to be correct/incorrect?":
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        "What are the limitations of the system?": {"answer": ""},
        "What kind of mistakes is the system likely to make?": {"answer": ""}
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    "How (global model-wide explanation)": {
        "How does the system make predictions?": {"answer": ""},
        "What features does the system consider?": {"answer": ""},
        "What is the system's overall logic?": {"answer": ""},
        "How does it weigh different features?": {"answer": ""},
        "What kind of rules does it follow?": {"answer": ""},
        "What are the top rules/features that determine its predictions?":
        ↪  {"answer": ""},
        "What kind of algorithm is used?": {"answer": ""},
        "How were the parameters set?": {"answer": ""}
    },
    "Others": {
        "How have updates affected the system?": {"answer": ""},
```

```
            "Does the system learn specifically from my interaction?": {"answer":
            ↪  ""},
            "Why is the system using or not using a given
            ↪  algorithm/feature/rule/dataset?": {"answer": ""},
            "What are the results of other people using the system?": {"answer":
            ↪  ""},
            "Who is responsible for this system? Who are the authors?": {"answer":
            ↪  ""}
        },
        "UI": {
            "What are the affordances of the system?": {"answer": ""},
            "What does this UI element do?": {"answer": ""}
        }
  }
}
```

ACF, Readability: Average Scores and Rank

Question	ACF		Readabillity	
	Score	Rank	Score	Rank
What kind of data was the system trained on?	0.840	5	13.61	9
What is the source of the training data?	0.909	1	15.71	24
How were the labels/ground-truth produced?	0.831	7	13.65	10
What is the sample size of the training data?	0.666	24	12.42	4
What dataset(s) is the system NOT using?	0.614	29	16.65	29
What are the potential limitations/biases of the data?	0.695	20	15.83	26
What kind of output does the system give?	0.528	31	14.58	15
What does the system output mean?	0.524	32	13.29	7
What is the scope of the system's capability? Can it do...?	0.548	30	14.82	16
How is the output used for other system component(s)?	0.679	23	16.77	31
How should I best utilize the output of the system?	0.648	26	17.38	33
How should the output fit in my workflow?	0.727	15	15.57	22
What is the source of the information object(s)?	0.719	17	15.39	19
What is the scope of the output data?	0.660	25	13.97	11
What is the amount of the output data?	0.633	28	16.75	30
What logic is used for the initial output/recommendation?	0.643	27	12.41	3
How accurate/precise/reliable are the predictions?	0.697	19	11.56	2
How often does the system make mistakes?	0.827	8	13.29	8
In what situations is the system likely to be correct/incorrect?	0.688	22	14.37	13
What are the limitations of the system?	0.496	34	14.45	14
What kind of mistakes is the system likely to make?	0.800	10	13.12	5
How does the system make predictions?	0.835	6	15.55	21
What features does the system consider?	0.776	11	19.14	35
What is the system's overall logic?	0.466	35	15.72	25
How does it weigh different features?	0.441	36	15.62	23
What kind of rules does it follow?	0.516	33	14.20	12
What are the top rules/features that determine its predictions?	0.847	4	17.75	34
What kind of algorithm is used?	0.889	2	16.50	27
How were the parameters set?	0.748	13	15.18	18
How have updates affected the system?	0.690	21	13.13	6
Does the system learn specifically from my interaction?	0.809	9	11.31	1
Why is the system using or not using a given algorithm/feature/rule/dataset?	0.746	14	16.54	28
What are the results of other people using the system?	0.723	16	15.48	20
Who is responsible for this system? Who are the authors?	0.868	3	14.93	17
What are the affordances of the system?	0.773	12	16.80	32
What does this UI element do?	0.717	18	19.79	36

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